Counterparts

Cian Dorr

Draft of 21st April 2015
Comments welcome

Contents

1 A-theories and B-theories **1**
- 1.1 Two key questions about time and modality 1
- 1.2 Propositions 13
- 1.3 Worlds and instants 24
- 1.4 Elitism and egalitarianism 32
- 1.5 Non-factualism and Relativism 38
- 1.6 Other A-theoretic themes 48
- 1.7 Summary and preview 53

2 Arguments about A- and B-theories **55**
- 2.1 Explicit quantification over times and worlds in semantics . . 55
- 2.2 Intervals versus instants 66
- 2.3 Utterance truth 66
- 2.4 Arguments from propositional anaphora 70
- 2.5 The psychological objection to the B-theories 80
- 2.6 The primordial objection to the B-theories 88
- 2.7 The argument from spacetime 94
- 2.8 Counterpart theory 109

3 Counterpart theory and the fixity of the qualitative **117**
- 3.1 Basic counterpart theory 117
- 3.2 Some puzzles for Lewis 124
- 3.3 Contingency-inducing quantification as *de re* restricted quantification 134
- 3.4 An exegetical interlude 142
- 3.5 Alternative accounts of contingency-inducing quantification . 147
- 3.6 Necessary and eternal existence 158

4 Counterpairings, worlds and instants 165
4.1 Multiply *de re* counterpart theory: two bad ideas 165
4.2 Three better ideas 173
4.3 Constraints on counterpairings 180
4.4 The necessity of identity 190
4.5 Worlds and instants in counterpart theory 199
4.6 Actuality 213
4.7 Counterpart theories as A-theories 216

5 Modal counterpart theory 229
5.1 The generality of modal counterpart theory 229
5.2 Island universes and constraints on counterparthood 238
5.3 Surprising dependencies 252
5.4 What is special about metaphysical modality? 272
5.5 Beyond counterpart theory 277

6 Temporal counterpart theory 282
6.1 Spacetime and the ontology of material objects 282
6.2 Fission and linearity 296
6.3 The modal profiles of instants 300
6.4 Time travel 300
6.5 Relativity <TK> 307

Bibliography 308

Chapter 4

Counterpairings, worlds and instants

4.1 Multiply *de re* counterpart theory: two bad ideas

Let us now turn to the general case of counterpart theory, where modal or temporal operators are applied to formulae that may contain more than one referential term. To begin with, let's see what happens if we just mechanically extend BMCT and BTCT in the most obvious way:

$$\text{BMCT+}\quad \begin{aligned} \Diamond\phi(t_1,\ldots,t_n) &:= \exists v_1 \ldots \exists v_n(\mathrm{C}v_1t_1 \wedge \cdots \wedge \mathrm{C}v_nt_n \wedge \phi(x_1,\ldots,x_n)) \\ \Box\phi(t_1,\ldots,t_n) &:= \forall v_1 \ldots \forall v_n((\mathrm{C}v_1t_1 \wedge \cdots \wedge \mathrm{C}v_nt_n) \rightarrow \phi(x_1,\ldots,x_n)) \end{aligned}$$

$$\text{BTCT+}\quad \begin{aligned} \mathsf{F}\phi(t_1,\ldots,t_n) &:= \exists v_1 \ldots \exists v_n(\mathrm{C_F}v_1t_1 \wedge \cdots \wedge \mathrm{C_F}v_nt_n \wedge \phi(x_1,\ldots,x_n)) \\ \mathsf{G}\phi(t_1,\ldots,t_n) &:= \forall v_1 \ldots \forall v_n((\mathrm{C_F}v_1t_1 \wedge \cdots \wedge \mathrm{C_F}v_nt_n) \rightarrow \phi(x_1,\ldots,x_n)) \\ \mathsf{P}\phi(t_1,\ldots,t_n) &:= \exists v_1 \ldots \exists v_n(\mathrm{C_P}v_1t_1 \wedge \cdots \wedge \mathrm{C_P}v_nt_n \wedge \phi(x_1,\ldots,x_n)) \\ \mathsf{H}\phi(t_1,\ldots,t_n) &:= \forall v_1 \ldots \forall v_n((\mathrm{C_P}v_1t_1 \wedge \cdots \wedge \mathrm{C_P}v_nt_n) \rightarrow \phi(x_1,\ldots,x_n)) \end{aligned}$$

Here $\phi(t_1,\ldots,t_n)$ is any formula in which the syntactically simple referring terms (including variables) are $t_1 \ldots t_n$, in order of first occurrence, and without repetition, and in which all other vocabulary is qualitative. '$\phi(v_1,\ldots,v_n)$' is the result of uniformly substituting for these terms variables $v_1 \ldots v_n$ which do not already appear in $\phi(t_1,\ldots,t_n)$.

BMCT+ and BTCT+ are hopeless. BMCT+ makes it far too easy for possibility claims to be true, and far too hard for necessity claims to be true; BTCT+ makes it far too easy for claims to the effect that something sometimes is the case to be true, and far too hard for claims to the effect that something is always the case to be true. Here are some examples that illustrate this point:

(i) Consider Socrates' unit set, Singleton-Socrates. We would like to say (at least in ordinary contexts) that necessarily, nothing other than Socrates is a member of Singleton-Socrates. But according to BMCT+, this is necessary only if for any counterparts x_1 and x_2 of Socrates and Singleton-Socrates, respectively, nothing other than x_1 is a member of x_2. It follows from this that if Singleton-Socrates has any nonempty counterparts, Socrates has at most one counterpart; for if he had several, each of them would have to be such that nothing other than it is a member of any of Singleton-Socrates's counterparts. And of course saying that Socrates has only one counterpart is not an option in any ordinary context, since we want to be able to say, for example, that Socrates could have died in battle and that Socrates could have died of old age. (Similarly, BTCT+ will not allow us to claim that Singleton-Socrates has never had any members other than Socrates.)

(ii) Let Phil be the proposition that Socrates is a philosopher. By the truth schema, necessarily, Phil is true iff Socrates is a philosopher. But according to BMCT+, this is necessary only if for any counterparts x_1 and x_2 of Socrates and Phil, x_2 is true iff x_1 is a philosopher. It follows from this that if Phil has any true counterparts, all of Socrates' counterparts are philosophers: if it is possible for Phil to be true, it is necessary that Socrates is a philosopher. Which is absurd: of course it is possible for Phil to be true, since is is possible for Socrates to be a philosopher; but it is not necessary (by ordinary contextual standards) that Socrates is a philosopher. (Similarly, BTCT+ will not allow us to claim that it has always been the case that Phil is true iff Socrates is a philosopher.)

(iii) Suppose we have a painted wooden statue, Athena. Some paint is part of Athena. (One does not need to deduct the weight of the paint from the reading on the scale when one is asked to determine Athena's weight.) But there is an object, Athena-Minus, that is made entirely of wood and overlaps all the wood in Athena throughout her life. It seems plausible that Athena-Minus could not possibly have been disjoint from Athena. But according to BMCT+, this claim is false so long as *any* counterpart of Athena-Minus is disjoint from *some* counterpart of Athena, which certainly true if Athena has any counterparts other than herself.

(iv) Similarly, BTCT+ entails that I have not always lived on Earth, so long as some past-counterpart of mine fails to live on some past-counterpart of Earth. And this is surely the case. Since I used to be a child and used to be a student, and earth Earth used to be populated by dinosaurs and used to be populated by wooly mammoths, both Earth and I have multiple past-counterparts. And it is false that each of my past-counterparts lives on *all* of Earth's past-counterparts, since I have never been living on many planets, and many of Earth's past-counterparts are planets.

(v) Consider the claim that there is exactly one person who could have been Prince Charles's mother. It seems plausible that this is true in many contexts, even when the quantifier 'exactly one' is taken as completely unrestricted. But according to BMCT+, this is equivalent to the claim that there is exactly one person who has a counterpart that is a mother of a counterpart of Charles. And this not plausible at all: in general, the mother of any counterpart of Charles will be the counterpart of many different people.

(vi) (v) has a temporal analogue which is even more compelling. It is obviously true that there is only one person, unrestrictedly speaking, who ever *has* been a mother of Prince Charles. But according to BTCT+, everyone who has the mother of a past-counterpart of Charles among their past-counterparts is someone who has been a mother of Charles. And on plausible assumptions, there are infinitely many such people: Elizabeth herself, as well as all (or all but one) of her temporal counterparts who are mothers. Similarly, BTCT+ entails that everyone who has a past-counterpart who lives in some past-counterpart of North America is someone who has lived in North America; again, there are probably infinitely many such people. Sider, the originator of temporal counterpart theory, notices this problem, and takes it to be so serious that he retreats to an ungainly hybrid view, on which 'person' has one use on which it applies to "stages" and a different use on which it applies to "worms", with temporal predications and operators presumably turning out ambiguous depending on whether the objects in question are stages or worms. I think a much better response to the problem is to give up BTCT+ in favour of some other analysis of multiply *de re* temporal claims. Of course, if we are going to take temporal counterpart theory seriously, we will have to get used to the claim that

> there are infinitely many people, unrestrictedly speaking. While this claim is counterintuitive, it is an inevitable commitment of any view that tries to reconcile the manifold picture with a view that avoids the primordial objection to the B-theory. The claim that there are infinitely many people is *much* easier to believe than the claim that there are infinitely many people who have lived in North America, let alone the claim that there are infinitely many people who have been mothers of Prince Charles.

Lewis's counterpart theory avoids some of the problems that beset BMCT+, by requiring all of the sequences of counterparts witnessing the truth of a given possibility-claim to belong to the same Lewis-world. While chapter 3 argued that in the case of singly *de re* claims, we do better to stick with BMCT rather than giving any distinctive role in the analysis to the concept of a Lewis-world, perhaps quantification over Lewis-worlds can come into its own in the case of multiply *de re* claims, as a way of co-ordinating the assignments of counterparts. The suggestion is that, for example, it is possible that Athena and Athena-Minus are disjoint iff Athena and Athena-Minus have disjoint counterparts *which are in the same Lewis-world*. In general:

LCT+
$$\begin{aligned}\Diamond\phi(t_1,\ldots,t_n) &:= \exists w v_1 \ldots v_n(Ww \wedge Iv_1w \wedge \cdots \wedge Iv_nw \\ &\qquad \wedge Cv_1t_1 \wedge \cdots \wedge Cv_nt_n \wedge \phi(v_1,\ldots,v_n)) \\ \Box\phi(t_1,\ldots,t_n) &:= \forall w v_1 \ldots v_n((Ww \wedge Iv_1w \wedge \cdots \wedge Iv_nw \\ &\qquad \wedge Cv_1t_1 \wedge \cdots \wedge Cv_nt_n) \rightarrow \phi(v_1,\ldots,v_n))\end{aligned}$$

This is close to LCT, except that I have simply left out the part where we add a restrictor 'in w' to each quantifier in $\phi(v_1,\ldots,v_n)$. As we saw in chapter 3, this is a bad idea if the quantifiers are supposed to be unrestricted (or qualitatively restricted), while if the quantifiers carry a *de re* restriction, we must make this explicit before applying any counterpart-theoretic analysis. Putting the restrictors back in would not help address any of the objections to LCT+ to be considered below.

LCT+ is an improvement on BMCT+ in one respect: it makes it easier for multiply *de re* necessity claims to be true, and harder for multiply *de re* possibility claims to be true. Just as 'Athena and Athena-Minus could not have been disjoint' gets to be true so long as no counterpart of Athena is disjoint from any counterpart of Athena-minus in the same Lewis-world, 'Necessarily, nothing other than Socrates is a member of Singleton-Socrates' gets to be be true provided that each nonempty counterpart of Singleton-Socrates which is in a Lewis-world has as its only member the unique counterpart of Socrates in that Lewis-world. Similarly, 'Necessarily, Phil is true iff Socrates is a philosopher' is true so long as each Lewis-world is such that either all of Socrates' counterparts in it are philosophers and all of Phil's counterparts in it are true, or none of Socrates' counterparts in it are philosophers and none of Phil's counterparts in it are true. (To get non-crazy results from LCT+, we had better find a way of understanding 'in' that allows sets and propositions to be 'in' Lewis-worlds.)

However—as Hazen (1979) argued, and as Lewis came to agree—LCT+ still makes it too hard for multiply *de re* necessity claims like these to be true. The problem is that according to Lewis, it sometimes happens that an object has multiple counterparts in the same Lewis-world. If any Lewis-world contains two counterparts of Socrates, it cannot be necessary that nothing other than Socrates is a member of Singleton-Socrates. If any Lewis-world contains two counterparts of Athena, then it cannot be necessary that Athena and Athena-Minus are not disjoint (unless it somehow manages to be the case that every counterpart of Athena-Minus in these worlds overlaps *all* of the counterparts of Athena in it, which is wildly implausible).

Why not avoid these problems by saying that no object has more than one counterpart in any Lewis-world? In CTQML, Lewis's reason for rejecting this suggestion arises from a certain analysis of counterparthood: 'x is a counterpart of y' is supposed to mean 'x is similar enough (in relevant respects) to y, and at least as similar to y as anything else in its Lewis-world'. If this were the only reason, we could reasonably brush the problem aside by saying 'So much the worse for that account of counterparthood!'. However, there are deeper reasons why Lewis is not in a position to stipulate that nothing has more than one counterpart in any Lewis-world. According to Lewis, some Lewis-worlds are symmetric—worlds of two-way eternal recurrence, for example. Moreover, some symmetric Lewis-worlds are, as I will put it, "qualitatively symmetric", in the sense that some nontrivial permutation on the set of their parts not only preserves all qualitative properties of and relations among the parts, but also preserves all qualitative relations between the parts and things in other Lewis-worlds.[1] Consider some qualitatively symmetric Lewis-world w. For any qualitative relation R, and object x not in w, if

[1]For Lewis, a sufficient condition for a symmetric world to be qualitatively symmetric is for none of its parts to bear any *perfectly natural* relation to anything that is not one of its parts.

x bears R to any part y of w, x also bears R to all the other qualitative duplicates of y in w. But 'counterpart' is supposed to express a qualitative relation, at least in many ordinary contexts: the facts about what qualitative properties a given object *could* have are fixed by the facts about what qualitative properties it *does* have.[2] So if an object has *any* counterpart in a qualitatively symmetric Lewis-world other than its own, it has many counterparts in that Lewis-world. The only way we could avoid claiming that some objects have multiple counterparts within the same Lewis-world, then, would be to say that objects in qualitatively symmetric Lewis-worlds are never counterparts of objects in any other Lewis-world. And that claim has the implausible consequence that it is never *possible* for an object to be part of a qualitatively symmetric Lewis-world with certain qualitative features unless that object is *in fact* part of a qualitatively symmetric Lewis-world with those qualitative features.[3]

This reasoning is, however, quite specific to Lewis's system. It does not arise in temporal counterpart theory: since the parts of spacetime ('Slices') that might play the role of Lewis-worlds in a temporal analogue of LCT+ are linked by a rich array of qualitative relations, there is no obvious reason why we could not impose a one-counterpart-per-Slice constraint if we wanted to.[4] And it might not arise in certain heterodox versions of modal counter-

[2]Actually, it wouldn't matter much if we considered non-qualitative counterpart relations, such as those that would be required to deal with an interpretation of 'possible' as 'consistent with Bill Clinton's knowledge'. For all of the objects which might plausibly figure as *de re* elements of the analysis of possibility claims made by *us* are spatiotemporally related to us. Thus qualitative relations between them, us, and objects in other strongly isolated Lewis-worlds are still invariant under the symmetries of those other Lewis-worlds.

[3]Biting the bullet here might be less implausible for someone who thought of counterparthood in general as involving some external relation of the sort that cannot hold between x and y unless some chain of nonlogical perfectly natural relations connects some part of x and some part of y (x and y are 'naturally linked', to use the terminology introduced below in section 5.2). This view makes it harder for ordinary possibility claims to be true: for an object to be possibly F, there needs to be a chain of perfectly natural relations connecting it to something that is F. Lewis's doctrine of Humean Supervenience, by contrast, commits him to thinking that ordinary modal judgments are consistent with the hypothesis that the only perfectly natural relations instantiated in our Lewis-world are spatiotemporal in character. This entails that nothing in our Lewis-world is naturally linked to anything in any other Lewis-world.

[4]Sider (1996) claims that the one-counterpart-per-Slice constraint fails in cases of fission and fusion, and argues that this gives the temporal counterpart theorist a more attractive account of such cases than anything available to "worm" theorists. Section 6.2 below will consider whether it is a good idea for temporal counterpart theorists to accept Sider's claims

part theory, e.g. versions which replace Lewis's ontology with some other expansive vision of reality (such as might be suggested by certain views in cosmology or in the foundations of quantum mechanics), or versions which replace Lewis's classical mereology account with some more finely articulated account of the structure of ordinary objects.

However, even if we were to endorse the claim that objects never have multiple counterparts in the same Lewis-world (or Slice), some of the problems for BMCT+ remain as problems for LCT+. Just like BMCT+, LCT+ entails that each person (in whatever Lewis-world) who has a counterpart that is the mother of a counterpart of Charles is someone who could have been Charles's mother. Since the relation of motherhood holds within a Lewis-world, the move from BMCT+ to LCT+ makes no difference in this case. Similarly, in the temporal case, the analogue of LCT+ using Slices will share BTCT+'s implausible consequences that there are infinitely many people who have been mothers to Charles, infinitely many people who have lived in North America, etc.

(Note that these last objections, which arise equally for LCT+ and BMCT+, essentially involve combining modal and temporal operators with unrestricted quantifiers. Those who embrace the "two languages" picture from CTQML might feel willing, on these grounds, to dismiss the claims in question as involving an illegitimate attempt to mix the two languages. But I have already explained, in section 3.3, why I don't think that such restrictions on intelligibility are tenable.)

Thus, LCT+ does at best a very partial job of overcoming the objections to BMCT+. Meanwhile, LCT+ introduces two very serious new problems which were not problems for BMCT+. First, so long as some Lewis-worlds contain no counterparts at all of certain objects, LCT+ will be unable to vindicate certain obviously valid principles of modal logic. Consider for example the following modal schemas:

(1) $\Diamond Fa \rightarrow \Diamond(Fa \lor Fb)$

(2) $\Box(Fa \land Fb) \rightarrow \Box Fa$

These seem as manifestly valid as anything could in the realm of modal logic. They are especially trivial instances of the closure of possibility and necessity under logical consequence. But look at the analysis of (1) given by

---

about these cases.

LCT+:

$$\exists w \exists x(Ww \wedge Ixw \wedge Cxa \wedge Fx) \rightarrow \\ \exists w \exists x \exists y(Ww \wedge Ixw \wedge Iyw \wedge Cxa \wedge Cyb \wedge (Fx \vee Fy))$$

In English: if some Lewis-world contains a counterpart of a that is F, then some Lewis-world contains both a counterpart of a and a counterpart of b, such that either the former or the latter is F. This is false if a has counterparts that are F, but all of them are in Lewis-worlds that contain no counterparts of b. Thus, given LCT+, guaranteeing the truth of all instances of (1) would require claiming that every object has at least one counterpart in every Lewis-world. (The same is easily seen to be true for (2).)

In the standard proof theory for modal logic, claims like (1) and (2) would be derived from the fundamental axiom K:

(K) $$\Box(\phi \rightarrow \psi) \rightarrow (\Box\phi \rightarrow \Box\psi)$$

together with the 'rule of necessitation', which says that $\Box\phi$ is a logical truth whenever ϕ is. Unsurprisingly, then, the cases where (1) and (2) fail according to LCT+ also turn out to involve failures of K. For example,

(3) $$\Box((Fa \wedge Fb) \rightarrow Fa) \rightarrow (\Box(Fa \wedge Fb) \rightarrow \Box Fa)$$

will have false instances if something lacks a counterpart in some Lewis-world, since its consequent is (2) while its antecedent a consequence of LCT+; .[5]

The failure of K is not simply a random bug in Lewis's analysis: it plays an important role in his defence of claims like 'Necessarily, Charles is human'. K plays a crucial role in the following argument against this claim:

(i) It is necessary that if Charles is human, then Elizabeth and Philip meet (premise).

(ii) It is not necessary that Elizabeth and Philip meet (premise).

(iii) If it is necessary that Charles is human, then it is necessary that Elizabeth and Philip meet (by K, from (i))

[5] As far as I know, the failure of Lewis's counterpart theory to validate K was first noted by Kripke in a letter to Lewis (Kripke 1969).

(iv) It is not necessary that Charles is human (by modus tollens, from (iii) and (ii)).

By rejecting K, one can accept both premises of this argument while rejecting its conclusion. However, this does not strike me as a compelling reason to reject K. Reasoning by K, and by the general principle that necessity and possibility are closed under logical consequence, is absolutely basic to the use of modal operators. In the case of the above argument, it seems better to appeal to the well-established context-sensitivity of modal operators than to give up this principle: (i), (ii), and 'It is necessary that Charles is human' may each be true in the contexts they most naturally evoke without all being true in the same context.

This problem with LCT+ could be avoided by requiring that every object have a counterpart in every Lewis-world. If the things 'in' a given Lewis-world are just the *parts* of that Lewis-world, this requirement would generate wildly implausible results. For example, consider a tiny Lewis-world w with no proper parts: w is in w, but nothing else is. If the parts of each Lewis-world had to include a counterpart for absolutely everything, w would have to be a counterpart of everything, making it true that everything could have been a spatiotemporally isolated mereological atom. This is surely not true in every legitimate context! And the broadenings of the 'in' relation which one might consider if, for example, one wanted to be able to think of *sets* as 'in' Lewis-worlds do not suggest any plausible way of getting it to be true that each object has a counterpart 'in' w.

The second of the new problems with LCT+ is more straightforward, and involves the T axiom. It is a straightforward consequence of LCT+ that for any two objects x and y, it is necessary that there is a Lewis-world containing both x and y. But if there are any two objects that are *not* in the same Lewis-world—and there had better be, otherwise LCT+ collapses into BMCT+—this is a violation of T. The objects in question are not worldmates, although it is necessary that they are. This is clearly unacceptable.

4.2 Three better ideas

I conclude that dragging Lewis-worlds, or anything else playing the logical role that Lewis-worlds play in LCT+, into the analyses of multiply *de re* modal and temporal claims is not the right strategy for the counterpart theorist. Fortunately, there are better options. One was developed by Lewis himself, in the 1983 postscripts to CTQML and in section 4.4 of OPW. According to this proposal, the analysis of a multiply *de re* modal claim involves a relation of counterparthood holding between *sequences* (n-tuples) of objects.[6] Here is a statement of this analysis, incorporating the insight that there is no need to include any special-purpose machinery for restricting quantifiers:

Sequences

$$\Diamond\phi(t_1,\ldots,t_n) := \exists v_1 \ldots \exists v_n(\mathrm{C}\langle v_1,\ldots,v_n\rangle\langle t_1,\ldots,t_n\rangle \wedge \phi(v_1,\ldots,v_n))$$
$$\Box\phi(t_1,\ldots,t_n) := \forall v_1 \ldots \forall v_n(\mathrm{C}\langle v_1,\ldots,v_n\rangle\langle t_1,\ldots,t_n\rangle \rightarrow \phi(v_1,\ldots,v_n))$$

Sequences (or at least, the necessitated biconditional corresponding to Sequences) actually follows from BMCT, together with the principle that that any n things form a sequence:

(4) $\Box\forall x_1 \ldots x_n \exists y(y = \langle x_1,\ldots,x_n\rangle)$

along with the following plausible principle of sequence-theoretic essentialism:

(5) $\forall x_1 \ldots x_n y(y = \langle x_1,\ldots,x_n\rangle \rightarrow \Box y = \langle x_1,\ldots,x_n\rangle)$.

It follows from (4) and (5) that a multiply *de re* possibility claim $\Diamond\phi(t_1,\ldots,t_n)$ is equivalent to a singly *de re* possibility claim about the sequence $s = \langle t_1,\ldots,t_n\rangle$, namely $\Diamond\exists y_1 \ldots y_n(s = \langle y_1,\ldots,y_n\rangle \wedge \phi(y_1,\ldots,y_n))$.[7] Given this equivalence, we can derive Sequences as follows:

$$\begin{aligned}\Diamond\phi(t_1,\ldots,t_n) &\leftrightarrow \Diamond\exists y_1 \ldots y_n(s = \langle y_1,\ldots,y_n\rangle \wedge \phi(y_1,\ldots,y_n))\\ &\leftrightarrow \exists z(\mathrm{C}sz \wedge \exists y_1 \ldots y_n(z = \langle y_1,\ldots,y_n\rangle \wedge \phi(y_1,\ldots,y_n)))\\ &\leftrightarrow \exists y_1,\ldots,y_n(\mathrm{C}\langle t_1,\ldots,t_n\rangle\langle y_1 \ldots,y_1\rangle \wedge \phi(y_1,\ldots,y_n))\end{aligned}$$

However, it is worth noting that Sequences does not *require* any doctrine of sequence-theoretic essentialism. Sequences even leaves us free to hold that

[6] In postscript D to the reprint of CTQML, Lewis presents the "counterparts of sequences" account not as a modification of the original translation from the formal modal language to the language of counterpart theory, but as an alternative to the obvious translation from English into the formal modal language. His idea is that the English sentence 'It is possible that Dee and Dum are unrelated' should be rendered formally as something like '$\exists z(z = \langle \text{Dee}, \text{Dum}\rangle) \wedge \Diamond\exists x\exists y(z = \langle x, y\rangle \wedge \text{Unrelated}(x, y))$' rather than '$\Diamond\text{Unrelated}(\text{Dee}, \text{Dum})$'. Given that the ultimate point is to come up with analyses that can be stated in English, I take it nothing turns on this distinction.

[7] Deriving this also requires NE, the principle that everything is necessarily identical to something; but as we have already seen in section 3.6, this is a consequence of BMCT.

the relation between an object x and its corresponding length-one sequence $\langle x\rangle$ is a contingent one: $\langle x, \langle x\rangle\rangle$ could have a counterpart $\langle y, z\rangle$ such that $z \neq \langle y\rangle$. If we do think this, then there will be a superficial conflict between SEQUENCES and BMCT: for a singly *de re* claim $\Diamond\phi(t)$, SEQUENCES yields the analysis $\exists y(\mathrm{C}\langle y\rangle\langle t\rangle \wedge \phi(y))$ while BMCT yields the analysis $\exists y(\mathrm{C}yt \wedge \phi(y))$. But of course, SEQUENCES entails that BMCT is true on *some* interpretation of 'counterpart', namely one on which 'x is a counterpart of y' means '$\mathrm{C}\langle x\rangle\langle y\rangle$', where C is the predicate of sequences used in SEQUENCES.

SEQUENCES is strictly more flexible than BMCT+ and LCT+. If we wanted to recover the effect of BMCT+ or LCT+ in some context, we could do so by analysing the counterpart relation on sequences in terms of a prior counterpart relation on objects, like this:

(6) $\mathrm{C}\langle x_1, \ldots, x_n\rangle\langle y_1, \ldots, y_n\rangle := \mathrm{C}x_1y_1 \wedge \ldots \wedge \mathrm{C}x_ny_n$

or like this

(7) $\mathrm{C}\langle x_1, \ldots, x_n\rangle\langle y_1, \ldots, y_n\rangle := \mathrm{C}x_1y_1 \wedge \ldots \wedge \mathrm{C}x_ny_n$, and $x_1 \ldots x_n$ are all in the same accessible Lewis-world.

But we have already seen why we should not expect to be able to do this in any ordinary context.[8]

By adopting appropriate postulates about the counterpart relation on sequences, SEQUENCES (and its temporal analogue) allows us to avoid all the problems that beset BMCT+:

(i) Provided that every counterpart of ⟨Socrates, Singleton-Socrates⟩ is an ordered pair $\langle x, \{x\}\rangle$ for some x, it is necessarily true (and always true) that Singleton-Socrates has no members other than Socrates.

(ii) Provided that every counterpart of ⟨Socrates, Phil⟩ is of the form ⟨x, the proposition that x is a philosopher⟩, it is necessarily true (and always true) that Phil is true iff Socrates is a philosopher.

[8] An alternative idea, due to Graeme Forbes (1985), is, roughly, to adopt SEQUENCES while analysing '$\mathrm{C}\langle x_1, \ldots, x_n\rangle\langle t_1, \ldots, t_n\rangle$' as 'for some world w, $\bigwedge_i$(either x_i is in w and $\mathrm{C}x_it_i$, or t_i has no counterpart in w and $x_i = t_i$)'. This analysis has the advantage of making (1) and (2) valid. However, it has implausible consequences as it stands, e.g., that if Gordon Brown is necessarily not an atomic Lewis-world, then for every x, necessarily, if x is an atomic Lewis-world, then Gordon Brown is Prime Minister. Forbes's remedy for this involves a much more radical departure from modal realist counterpart theory, in which atomic predicates get endowed with extra argument places that in effect turn them into relations to worlds.

(iii) Provided that $\langle$Athena, Athena-Minus$\rangle$ has no counterpart that is an ordered pair of two disjoint items, it is impossible for Athena and Athena-Minus to be disjoint.

(iv) Provided that $\langle$me, Earth$\rangle$ has no past-counterpart whose first element fails to live on its second element, it has always been the case that I have lived on Earth.

(v) Provided that there is no x other than Elizabeth such that $\langle x$, Charles$\rangle$ has a counterpart whose first element is the mother of its second element, Elizabeth is the only thing which could have been a mother of Charles. And The mere fact that some object x has Elizabeth as a counterpart gives us no special reason to think that $\langle x$, Charles$\rangle$ has $\langle$Elizabeth, Charles$\rangle$, or any other ordered pair whose first element is the mother of its second element, among its counterparts. (We might plausibly suppose that typically, when the elements of some ordered pair are in different Lewis-worlds, the elements of the counterparts of that pair are also in different Lewis-worlds.)

(vi) Similarly, provided that there is no x other than Elizabeth such that $\langle x$, Charles$\rangle$ has a past-counterpart whose first element is the mother of its second element, Elizabeth is the only thing which has ever been a mother of Charles. And provided that there are only finitely many people x such that $\langle x$, North America$\rangle$ has a past-counterpart whose first element lives in its second element, there are only finitely many people that have ever lived in North America. (If we have some notion of an object being "in" a Slice, such that the temporal counterparts of an objects are generally "in" different Slices, then we might plausibly suppose that ordered pairs whose elements are "in" different Slices only have other such ordered pairs as past- and future-counterparts.[9])

Nor is SEQUENCES subject to any of the distinctive problems which beset LCT+. True, it does not validate the modal axioms K and T all by itself. However, there are further constraints which we could impose on the counterpart relation for sequences which would have the effect of validating those axioms without having any further untoward effects. Validating T is straightforward: we need only require that every sequence has itself as a counterpart (even if its elements belong to different Lewis-worlds). It is not

[9]But perhaps an exception should be made in worlds with time-travel: see section 6.4.

quite so easy to write down constraints which have the effect of validating K, although as we will see in section 4.3, it can be done. In particular cases, it is easy to see what is needed to guarantee the closure of possibility and necessity under some logically valid inference. For example, to validate the inference from 'Possibly *Fa* and *Gb*' to 'Possibly *Gb* and *Fa*', we can say that whenever $\langle x_1, x_2\rangle$ is a counterpart of $\langle y_1, y_2\rangle$, $\langle x_2, x_1\rangle$ is a counterpart of $\langle y_2, y_1\rangle$. Similarly, to validate the inference from 'Possibly *Fa* and *Gb*' to 'Possibly *Fa*', we can say that whenever $\langle x_1, x_2\rangle$ is a counterpart of $\langle y_1, y_2\rangle$, $\langle x_1\rangle$ is a counterpart of $\langle y_1\rangle$. These cases illustrate the following plausible more general constraint: when we have a pair of counterpart sequences, then any sequence of elements of the first has the corresponding sequence of elements of the second as a counterpart. To state this idea rigorously, it will be convenient to identify sequences of length n with functions whose domain is the first n natural numbers. Then we can put the claim like this:

Permutation: If π is a function from the natural numbers $< m$ to the natural numbers $< n$, and σ_1 and σ_2 are sequences of length n such that σ_2 is a counterpart of σ_1, then $\sigma_2 \circ \pi$ is a counterpart of $\sigma_1 \circ \pi$.

Permutation suggests that in taking 'counterpart' to apply to *pairs of sequences*, we are importing superfluous structure. The order of the elements of the sequences is irrelevant except insofar as it sets up a correspondence between the members of the first sequence and the members of the second sequence. Thus the representational work of the pair of sequences $\langle\langle a_1, \ldots, a_n\rangle, \langle b_1, \ldots, b_n\rangle\rangle$ could be done more perspicuously by the corresponding set of ordered pairs: $\{\langle a_1, b_1\rangle, \ldots, \langle a_n, b_n\rangle\}$. If we modify Sequences in accordance with this idea, we will want to replace the two-place predicate of sequences C with a one-place predicate '**C**' applying to sets of ordered pairs, which I will pronounce 'is a counterpairing'.

Insofar as it relies on quantification over *sets*, this proposal is still not ideal: we can achieve some desirable generality by eliminating quantification over sets in favour of plural quantification. (Following Boolos (1985), many philosophers, conspicuously including Lewis, hold that such quantification is not to be understood in terms of first-order quantification over sets or anything else.) Thus officially, 'counterpairing' will be a *plural* predicate, applying to ordered pairs taken collectively.[10] It will be convenient to state

[10] Having said this, I will sometimes allow myself to be sloppy, saying things like 'counterpairings are relations'.

the analysis in the notation of second-order logic, using 'Xx' to abbreviate 'x is one of X', and 'Rxy' to abbreviate '$\langle x, y\rangle$ is one of R'. So we have:

Counterpairings

$$\Diamond\phi(t_1,\ldots,t_n) := \exists R \exists v_1 \ldots \exists v_n(\mathbf{C}R \wedge Rt_1v_1 \wedge \ldots \wedge Rt_nv_n \wedge \phi(v_1,\ldots,v_n))$$

$$\Box\phi(t_1,\ldots,t_n) := \forall R \forall v_1 \ldots \forall v_n((\mathbf{C}R \wedge Rt_1v_1 \wedge \ldots \wedge Rt_nv_n) \rightarrow \phi(v_1,\ldots,v_n))$$

If we prefer to work with Sequences, we can recover it from Counterpairings by adopting the following analysis:

(8) $\mathrm{C}\langle x_1,\ldots,x_n\rangle\langle y_1,\ldots,y_n\rangle := \exists R(\mathbf{C}R \wedge Rx_1y_1 \wedge \cdots \wedge Rx_ny_n)$

And for the special case of singly *de re* sentences, we can of course recover BMCT by analysing 'x is a counterpart of y' as '$\exists R(\mathbf{C}R \wedge Rxy)$'.

Like Sequences, Counterpairings needs to be supplemented with further axioms if we want to recover a reasonable modal logic. Section 4.3 will consider in detail how such axioms about 'counterpairing' map on to schemas involving modal operators.

From my point of view, Counterpairings has one big advantage over Sequences, namely the fact that it can very easily be extended to cover sentences in which *plural* terms and free variables occur in the scope of modal operators. All we need to do is to define 'RXY' to abbreviate the following conjunction: (i) for any x that is one of X, there is some y that is one of Y such that Rxy and (ii) for any y that is one of Y, there is some x that is one of X such that Rxy. We can then use Counterpairings as it stands to analyse modal sentences involving plural terms: we simply allow the terms $t_1,\ldots,t_n$ to be plural as well as singular, understanding that when t_i is plural, its corresponding variable v_i will also be plural.

I have focused here on the modal case, but of course Counterpairings has a natural temporal analogue, in which 'counterpairing' will be replaced with 'future-counterpairing' or 'past-counterpairing' as appropriate.

It is worth mentioning a third proposal for dealing with multiply *de re*, which works only for temporal counterpart theory. Prior (1967) introduced a *metrical* tense operator $\mathsf{F}n$, pronounced 'it will be the case n units of time hence that...' (or when n is negative, 'it was the case $-n$ units of time ago that...'). If we can make sense of this operator, we can plausibly analyse the usual tense operators as the result of quantifying into the 'n' position:

$\mathsf{F}\phi$ and $\mathsf{P}\phi$ are respectively equivalent to $\exists n(n > 0 \wedge \mathsf{F}n\phi)$ and $\exists n(n < 0 \wedge \mathsf{F}n\phi)$. A temporal counterpart theorist will naturally want to analyse the metric tense operator in terms of a corresponding metric notion of temporal counterparthood, which lets us say that x is an 'n-unit temporal counterpart' of y; for short, $Cnxy$. For the case of the singly *de re*, there are two obvious ways of implementing this, depending on whether we use universal or existential quantification:

$$\mathsf{F}n\phi(t) := \exists x(Cnxt \wedge \phi(x))$$
$$\mathsf{F}n\phi(t) := \forall x(Cnxt \rightarrow \phi(x))$$

Which of these should we choose? There would be no need to choose we adopted the following 'uniqueness postulate': for every number n and every object x, necessarily, x has *exactly one* n-unit temporal counterpart. And the uniqueness postulate could be justified on the grounds that it accounts for the validity of the following schemas for $\mathsf{F}n$:

$$\mathsf{F}n\phi \wedge \mathsf{F}n\psi \rightarrow \mathsf{F}n(\phi \wedge \psi)$$
$$\mathsf{F}n(\phi \vee \psi) \rightarrow \mathsf{F}n\phi \vee \mathsf{F}n\psi$$
$$\mathsf{F}n\neg\phi \rightarrow \neg\mathsf{F}n\phi$$

If we do adopt the uniqueness postulate, then it is a straightforward matter to extend our analysis to the case of the multiply *de re*, as follows:

$$\mathsf{F}n\phi(t_1,\ldots,t_m) := \exists v_1\ldots\exists v_m(Cnv_1t_1 \wedge \cdots \wedge Cnv_mt_m \wedge \phi(v_1\ldots v_m))$$

or (equivalently, given the uniqueness postulate):

$$\mathsf{F}n\phi(t_1,\ldots,t_m) := \forall v_1\ldots\forall v_m((Cnv_1t_1 \wedge \cdots \wedge Cnv_mt_m) \rightarrow \phi(v_1\ldots v_m)).$$

The objections to BMCT+ all involves cases where objects have multiple counterparts, or where the same object is a counterpart of multiple items. Given the uniqueness postulate and the plausible further assumption that $Cnxy$ is equivalent to $C{-}nyx$, this never happens for the 'n-unit temporal counterpart' relation. So we can analyse $\mathsf{F}n$ as above, and then follow Prior's analysis of the ordinary tense operators in terms of $\mathsf{F}n$.

This analysis need not be thought of as a competitor to Counterpairings. If we analyse 'R is a counterpairing' as $\exists n\forall x\forall y(Rxy \leftrightarrow Cnyx)$, we can recover Counterpairings *modulo* the uniqueness postulate. However, Counterpairings is less committal than the approach using the metric tense operator. First, it does not require the uniqueness postulate. And second, it does not require us to make sense of the question how many units of time apart a given pair of temporal counterparts are—a task which may not be straightforward, especially in the setting of relativistic spacetime.

4.3 Constraints on counterpairings

Since modal operators are context-dependent, the answer to the question what it is for some ordered pairs to be a counterpairing must vary from one context to another. But it is plausible that certain schematic principles involving modal operators are valid across all contexts. These will correspond to constraints which all legitimate readings of 'counterpairing' must meet. Similarly, while there may be some context-sensitivity in temporal operators (see chapter 6), some schematic principles involving them are valid, and these will correspond to constraints which all legitimate readings of 'future-counterpairing' and 'past-counterpairing' must meet.

Let us consider the modal case first, beginning with the central axiom, K:

K $$\Box(\phi \to \psi) \to (\Box\phi \to \Box\psi).$$

K does not follow from Counterpairings by itself. For example,

$$\Box((Fa \wedge b = b) \to Fa) \to (\Box(Fa \wedge b = b) \to \Box Fa)$$

is false if some counterpairings map a to objects that are not F, but none of these counterpairings have b in their domain.

The most straightforward way to guarantee the truth of every instance of K is to lay it down as an axiom, true across all contexts, that *counterpairings have universal domains*. That is:

(9) $$\forall R(\mathbf{C}R \to \forall x \exists y Rxy)$$

It is easy to see that this will validate K.

We don't, strictly speaking, need something as strong as this to validate K. It would be enough if we stipulated that every counterpairing can be *extended* to a counterpairing with a universal domain:

(10) $$\mathbf{C}R \rightarrow \exists S(\mathbf{C}S \wedge \forall x \forall y(Rxy \rightarrow Sxy) \wedge \forall x \exists y Sxy).$$

But if K is valid, we lose nothing by restricting our attention to those counterpairings that do have universal domains: if any counterpairing witnesses the truth of '$\Diamond\phi(t_1, \ldots, t_n)$', some counterpairing whose domain is universal does. The reasoning is simple. Let U be *all the things there are*—the things such that everything is one of them. By the plural version of COUNTERPAIRINGS, it is (trivially) necessary that if $\phi(t_1, \ldots, t_n)$ then $\phi(t_1, \ldots, t_n) \wedge \forall x(Ux \rightarrow Ux)$. Suppose that it is possible that $\phi(t_1, \ldots, t_n)$. Then by K, it is possible that $\phi(t_1, \ldots, t_n) \wedge \forall x(Ux \rightarrow Ux)$. By COUNTERPAIRINGS, this is true iff there is a counterpairing R and objects $v_1, \ldots, v_n, X$ such that $Rt_1v_1 \ldots$ and Rt_nv_n and RUX and $\phi(v_1, \ldots, v_m)$ and $\forall x(Xx \rightarrow Xx)$. But for RUX to be true, R must have a universal domain. So if it is possible that $\phi(t_1, \ldots, t_n)$, there must be a counterpairing R with universal domain, and objects $v_1, \ldots, v_n$ such that $Rt_1v_1 \ldots$ and Rt_nv_n and $\phi(v_1, \ldots, v_n)$. Thus, even if we started with a sense of 'counterpairing' that applied to some relations that don't have universal domains, we will end up with equivalent analyses if we redefine 'counterpart' so that it only applies to relations with universal domains.

(We can imagine *other* analyses of modal operators in terms of counterpairings on which the existence of counterpairings with non-universal domains makes a real difference. For example, there might be an analysis under which 'I could have failed to be identical to anything' is equivalent to 'There is a counterpairing whose domain does not include me'. But as we saw in section 3.6, there are formidable challenges facing reductionists about modality who want to recover an intuitive account of the possibility of nonexistence. While the idea of using counterpairings whose domain excludes x to represent possibilities of x's not being anything is initially attractive, it is very hard to see how such an analysis could allow us to endorse claims like 'Possibly, I am not identical to anything but still could have been a philosopher', or how its temporal analogue could allow us endorse claims like 'It will be the case that (I am not identical to anything but still used to be a philosopher)'. Constructing a theory consistent with such claims would require a major departure from COUNTERPAIRINGS.)

Another axiom that is valid in all standard quantified modal logics is Leibniz's Law:

(LL) $x = y \rightarrow (\phi \rightarrow \psi)$

where ψ is derived from ϕ by replacing one or more occurrences of x with occurrences of y. An instance of LL is $x = y \rightarrow (\Box x = x \rightarrow \Box x = y)$. Since $\Box x = x$ is a theorem, this simplifies to the principle that identity is necessary:

(NI) $x = y \rightarrow \Box x = y$.

Given Counterpairings, the universal closure of NI is equivalent to the claim that *counterpairings are functional*: whenever R is a counterpairing and Rxy and Rxz, $y = z$. A straightforward induction on the complexity of formulas shows that this condition suffices for the full generality of LL in the language of quantified modal logic.

LL seems about as obviously valid as any schema could. And the obviousness persists when we consider particular instances of LL, including the instance required for the derivation of NI. If x is necessarily identical to x, and y *is* x, then how could y fail to be necessarily identical to x? Nevertheless, LL has been controversial in the context of counterpart theory: Lewis rejected it in CTQML, and many philosophers think of the failure of LL as one of the central interesting things about counterpart theory. In section 4.4 below I will consider Lewis's reasons for rejecting LL, and argue that they are not persuasive.

Along with LL, the requirement that counterpairings be functional also validates a plausible principle of plural modal logic, according to which when a thing is one of some things, it is necessarily one of them:

(NO) $Xx \rightarrow \Box Xx$.

Suppose x is one of X, and R is a counterpairing such that Rxy and RXY. Part of what it means to say that RXY is that for every x' that is one of X, there is some y' that is one of Y such that $Rx'y'$. But if counterpairings are functional, then y is the only thing such that Rxy; thus y must be one of Y. In fact, the universal generalisation of NO is *equivalent* to the claim that all counterpairings are functional. For if some counterpairing R maps some object x to two distinct objects y_1 and y_2, we could take X to be the objects identical to x (following standard philosophical usage, I am allowing that "some things" are such that such that there is only one of them), and Y to be the objects identical to y_1. Then RXY and Rxy_2 even though y_2 is not one of Y; so it is not necessary that Xx.This is unsurprising, given that such counterpairings generate counterexamples to the necessity of identity: if one thought, concerning Russell, that Russell and he are identical but could have failed to be, one would also presumably think, concerning Russell and Whitehead, that Russell is one of them but could have failed to be.

If one rejected NE, one might be worried about NO even if one had no doubts about the necessity of identity. Consider Russell and Whitehead again. Would Russell still have been one of them if Whitehead had failed to be anything at all? Would Russell still have been one of them if *Russell* had failed to be anything at all? The answers are not obvious. If we tried to tinker with counterpart theory to allow for contingent existence, we would face delicate and difficult questions about how our analysis should deal with occurrences of 'is one of' in modal contexts. But if we follow the advice of section 3.6 and don't bother, I can't think of any other reason to worry about the consequence that NO and NI stand or fall together.

The combination of the two constraints we have considered so far—that all counterpairings have universal domains, and that all counterpairings are functional—would allow us to simplify the formulation of Counterpairings by employing a function-symbol notation. Let us use '$R[x]$' to stand for 'the object y such that Rxy', and likewise '$R[X]$' to stand for 'the objects Y such that RXY'. In this notation, Counterpairings looks like this:

Counterpairings*
$$\Diamond\phi(t_1,\ldots,t_n) := \exists R(\mathbf{C}R \wedge \phi(R[t_1],\ldots,R[t_n]))$$
$$\Box\phi(t_1,\ldots,t_n) := \forall R(\mathbf{C}R \rightarrow \phi(R[t_1],\ldots,R[t_n]))$$

However, this notation creates the potential for scope ambiguity when dealing with iterated modalities. If we want Counterpairings* to be equivalent to Counterpairings, we must be careful to treat the complex terms $R[t_i]$ as having *widest possible scope*. We want '$\Diamond\Diamond F(a)$' to be equivalent to '$\exists R(\mathbf{C}R \wedge \exists S(\mathbf{C}S \wedge F(S[R[a]])))$', not '$\exists R(\mathbf{C}R \wedge \exists S(\mathbf{C}S \wedge F(S[R][S[a]])))$'. One rigorous way to achieve this would be to follow Russell in taking sentences involving functor-notation as abbreviations of quantified sentences in a language free of such notation, and stipulate that the Russellian translations are to be applied in such a way that the quantifier introduced by each complex term takes the widest possible scope.

There are many other formal schemas which have been thought to be logically valid for some or all readings of 'necessarily' and 'possibly'. Many of these correspond naturally to constraints on counterpairings. Several of these schemas, including those we have already considered, are listed in Table 4.1, together with the constraints on counterpairings that correspond to them. Some technical remarks on these results:

(i) Standard modal logics are closed under the rule of necessitation: if ϕ is a theorem, so is $\Box\phi$. If all contingency is *de re* contingency, any theory whose axioms are all qualitative is guaranteed to have this feature. More generally, a theory will be closed under necessitation whenever its theorems are indifferent to the identities of particular objects: if $\phi(t_1, \ldots, t_n)$ is a theorem, $\forall v_1 \ldots \forall v_n \phi(v_1, \ldots, v_n)$ is a theorem too. Assuming that 'counterpairing' is itself a qualitative predicate, this means we will get necessitation for free, since the kinds of constraints we are currently considering are *general* truths about what is required for a plurality of ordered pairs to be a counterpairing.

(ii) The axioms listed stand in the following logical relations: (a) T entails D; (b) Given K and D, 5 and T are jointly equivalent to 4 and B; (c) Given K and B, 4 is equivalent to 5, ND is equivalent to NI, NNO is equivalent to NO, and BF is equivalent to CBF. The same entailments can easily be seen to hold among the corresponding constraints on counterpairings.

(iii) Given Counterpairings (understood so as to apply to plural as well as singular terms), BF2 entails BF, and the same entailment holds between the corresponding constraints.

(iv) Given Counterpairings, the constraints listed in the right-hand column entail every instance of the corresponding axiom-schemas in the left-hand column, with the proviso that for 4, *B* and 5, the entailments only hold on the assumption that 'counterpairing' is a qualitative predicate.[11] In proving that these entailments hold, we need to rely on the fact that our definition of 'RXY' is well-behaved, in the following ways: if R is functional ($y = z$ whenever Rxy and Rxz), then it is also "functional on plural arguments": if RXY and RXZ, then $\forall x(Yx \leftrightarrow Zx)$; similarly for the property of being one-to-one. If S is the converse of R

[11] Our analysis allows non-qualitative senses of 'counterpairing', corresponding to contexts where 'possible' means something like 'historically possible as of 1900' or 'consistent with Clinton's beliefs in 1995'. For these, something a bit more complicated is needed to validate 4, *B* or 5. Take for instance a notion of 'counterpairing' that takes one extra singular argument. For 4 ('$\Box_x\phi \rightarrow \Box_x\Box_x\phi$') to be valid for the corresponding sense of necessity, we need it to be the case that whenever R is an x-counterpairing, S is a y-counterpairing, and Rxy, $S \circ R$ is an x-counterpairing.

Axiom		*Constraint*
Duality	$\Box\neg\phi \leftrightarrow \neg\Diamond\phi$	Automatic from Counterpairings
K	$\Box(\phi \to \psi) \to (\Box\phi \to \Box\psi)$	Counterpairings have universal domains: $\mathbf{C}R \to \forall x \exists y Rxy$
D	$\Box\phi \to \Diamond\phi$	There is at least one counterpairing: $\exists R\mathbf{C}R$
T	$\Box\phi \to \phi$	The identity relation is a counterpairing: $\forall x \forall y(Rxy \leftrightarrow x = y) \to \mathbf{C}R$
B	$\phi \to \Box\Diamond\phi$	R^{-1} (the converse of R) is a counterpairing when R is: $(\mathbf{C}R \wedge \forall x \forall y(Sxy \leftrightarrow Ryx)) \to \mathbf{C}S$
4	$\Box\phi \to \Box\Box\phi$	$R \circ S$ (the relative product of R with S) is a counterpairing when R and S are: $(\mathbf{C}R \wedge \mathbf{C}S \wedge \forall x \forall y(Qxy \leftrightarrow \exists z(Sxz \wedge Rzy))) \to \mathbf{C}Q$
5	$\Diamond\phi \to \Box\Diamond\phi$	$R \circ S^{-1}$ is a counterpairing when R and S are: $(\mathbf{C}R \wedge \mathbf{C}S \wedge \forall x \forall y(Qxy \leftrightarrow \exists z(Szx \wedge Rzy))) \to \mathbf{C}Q$
NI NO	$x = y \to \Box x = y$ $Xx \to \Box Xx$	Counterpairings are functional: $(\mathbf{C}R \wedge Rxy \wedge Rxz) \to y = z$
ND NNO	$x \neq y \to \Box x \neq y$ $\neg Xx \to \Box\neg Xx$	Counterpairings are one-one: $(\mathbf{C}R \wedge Rxz \wedge Ryz) \to x = y$
CBF CBF2	$\Box\forall x\phi \to \forall x\Box\phi$ $\Box\forall X\phi \to \forall X\Box\phi$	Automatic from Counterpairings
BF	$\forall x\Box\phi \to \Box\forall x\phi$	Every counterpairing can be extended so that any given object is in its range: $\mathbf{C}R \to \forall x \exists S(\mathbf{C}S \wedge \forall z_1 \forall z_2(Rz_1z_2 \to Sz_1z_2) \wedge \exists y Syx)$
BF2	$\forall X\Box\phi \to \Box\forall X\phi$	Counterpairings have universal ranges: $\mathbf{C}R \to \forall y \exists x Rxy$

Table 4.1. Some modal axioms, together with constraints on counterpairings that generate them.

($\forall x \forall y(Sxy \leftrightarrow Ryx)$), then SXY iff RYX. And if Q is the relative product of R and S ($\forall x \forall y(Qxy \leftrightarrow \exists z(Rxz \wedge Szy))$), then QXY iff $\exists Z(RXZ \wedge SZY)$.[12]

(v) With the exception of NI/NO, ND/NNO, and D, the modal axiom-schemas do not actually entail the corresponding constraints. Rather, the relation is the one we have already seen in the case of *K*: if we have a sense of 'counterpairing' under which all instances of the axiom-schema are true (given Counterpairings), we can use it to define a new sense of 'counterpairing' which satisfies the corresponding constraint; and the analyses we get when we use the new, defined sense of 'counterpairing' will be logically equivalent to the originals. In the case of K, BF and BF2, the new sense will *strengthen* the old sense just enough to satisfy the constraint—for BF2, for example, we can define a counterpairing in the new sense to be a counterpairing in the old sense whose range is universal. In the case of T, B, 4 and 5, the new sense will *weaken* the old sense just enough to satisfy the constraint. For 4, for example, the new sense is the closure of the old sense under the operation of taking relative products.[13] We can prove that the analyses we get from Counterpairings using the new sense of 'counterpairing'

[12] I will prove the last of these facts, and show how to use it to establish the entailment for S4. *Right-to-left*: suppose that RXZ and SZY. Then $\forall x{\in}X\ \exists z{\in}Z(Rxz)$ and $\forall z{\in}Z\ \exists y{\in}Y(Szy)$, so $\forall x \in X\ \exists y \in Y\ \exists z(Rxz \wedge Szy)$. So $\forall x \in X\ \exists y \in Y(Qxy))$, since $Q = S \circ R$. By analogous reasoning, $\forall y{\in}Y\ \exists x{\in}X(Qxy))$; hence QXY. *Left-to-right*: Let Z comprise all and only those objects z such that for some x and y, $Xx \wedge Yy \wedge Rxz \wedge Szy$. If QXY, then $\forall x{\in}X\ \exists y{\in}Y\ Qxy$. So $\forall x \in X\ \exists y \in Y\ \exists z(Rxz \wedge Szy)))$. By the definition of Z, every such z is one of Z. So $\forall x{\in}X\ \exists z{\in}Z(Rxz)$. And it follows from the definition of Z that $\forall z{\in}Z\ \exists x{\in}X(Rxz)$. Hence RXZ. Analogous reasoning shows that SZY.

We can now show that if $R \circ S$ is a counterpairing whenever R and S are, and $\Box\phi(t_1, \ldots, t_n)$, then $\Box\Box\phi(t_1, \ldots, t_n)$. By Counterpairings, if $\Box\phi(t_1, \ldots, t_n)$, then $\phi(v_1, \ldots, v_n)$ whenever Q is a counterpairing and $Qt_1v_1 \ldots$ and Qt_nv_n. By the constraint, it follows that this is also true whenever $Q = R \circ S$ and R and S are counterpairings. By the above lemma, $Qt_iv_i \leftrightarrow \exists u(St_iu \wedge Ruv_i)$, whether t_i, v_i and u are singular or plural. So $\phi(v_1, \ldots, v_n)$ whenever R and S are counterpairings and $\exists u_1 \ldots \exists u_n\ St_1u_1 \wedge Ru_1v_1 \wedge \cdots \wedge St_nu_n \wedge Ru_nv_n$. (Here u_i and v_i are plural variables iff t_i is a plural term.) That is: whenever S is a counterpairing and $St_1u_1 \wedge \cdots \wedge St_nu_n$, then: whenever R is a counterpairing and $Ru_1v_1 \wedge \cdots \wedge Ru_nv_n$, then: $\phi(v_1, \ldots, v_n)$. So by a double application of Counterpairings, $\Box\Box\phi(t_1, \ldots, t_n)$.

[13] This closure can be defined with a little trickery using plural quantification over ordered triples. Thus: R is a counterpairing in the extended sense := there is a number n and some ordered triples X such that (i) the first element of each of X is a number less than n; (ii) whenever $0 \leq i < n$, the ordered pairs $\langle x, y\rangle$ such that $\langle i, x, y\rangle$ is one of X are a counterpairing in the old sense; (iii) any ordered pair $\langle x, y\rangle$ is one of R iff for some function f, $f(0) = x$ and $f(n) = y$ and whenever $0 \leq i < n$, $\langle i, f(i), f(i+1)\rangle$ is one of X.

will be equivalent to analyses using the old sense of 'counterpairing'.[14]

The situation in the case of temporal counterpart theory is similar: given the temporal version of Counterpairings, we can write down constraints involving the predicates 'future-counterpairing' and 'past-counterpairing' that correspond to various standard candidate axioms for tense logic. Each of the candidate modal axioms in Table 4.1 corresponds to two different tense-logical axioms, one derived by replacing $\Box$ and $\Diamond$ with H and P, the other derived by replacing $\Box$ and $\Diamond$ with G and F. I will refer to these by subscripting the corresponding modal axiom with 'F' or 'P': thus $\mathrm{K_F}$ is the schema '$\mathsf{G}(\phi \to \psi) \to (\mathsf{G}\phi \to \mathsf{G}\psi)$', and 4_P is the schema '$\mathsf{H}\phi \to \mathsf{HH}\phi$'. (Note however that there is no temptation to regard the temporal versions of T, B, or 5 as valid.) Some further candidate tense-logical axioms are listed in Table 4.2, together with corresponding constraints on temporal counterpairings.

Some remarks:

(i) The relations between the columns are the same as in the modal case. In each case, the claim in the second column together with (the temporal analogue of) Counterpairings entails the schema in the first column. In the other direction, if we have an interpretation of 'temporal counterpairing', 'future-counterpairing' and 'past-counterpairing' which renders every instance of the schema true (given Counterpairings), we can use them to define new interpretations which yield logically equivalent analyses.

[14] Here is the proof for B: the others are similar. Let $\mathbf{C}^*R$ be defined as $\mathbf{C}R \lor \mathbf{C}R^{-1}$; let $\Diamond^*$ and $\Box^*$ be defined as in Counterpairings but using $\mathbf{C}^*$ instead of $\mathbf{C}$. It is obvious from the definition that for any formula ϕ, $\Diamond\phi$ entails $\Diamond^*\phi$ and $\Box^*\phi$ entails $\Box\phi$; what we need to show is that the converse entailments also hold. For simplicity, suppose ϕ contains just one singular term t and no plural terms. Assume $\Diamond^*\phi(t)$: that is, $\exists R \exists x((\mathbf{C}R \lor \mathbf{C}R^{-1}) \land Rtx \land \phi(x))$. If $\mathbf{C}R$, then evidently $\Diamond\phi(t)$; so suppose $\mathbf{C}R^{-1} \land Rtx \land \phi(x)$. Since B is valid for $\Box$ and $\Diamond$, it follows from $\phi(x)$ that $\Box\Diamond\phi(x)$: that is,

$$\forall R_1 \forall z_1((\mathbf{C}R_1 \land R_1 y z_1) \to \exists R_2 \exists z_2(\mathbf{C}R_2 \land R_2 z_1 z_2 \land \phi(z_2)))$$

Putting R^{-1} for R_1 and t for z_1, we can deduce that

$$\exists R_2 \exists z_2(\mathbf{C}R_2 \land R_2 t z_2 \land \phi(z_2))$$

—that is, $\Diamond\phi(t)$. Similar reasoning shows that $\Box\phi(t)$ entails $\Box^*\phi(t)$. The extension to sentences containing multiple variables is straightforward. For the extension to plural variables, we also need the fact that if $S = R^{-1}$ then SXY iff RYX; this follows from the definition of SXY.

Axiom		*Constraint*
Mixing_P	$\phi \to \mathsf{HF}\phi$	R is a past-counterpairing $\to$ R^{-1} is a future-counterpairing
Mixing_F	$\phi \to \mathsf{GP}\phi$	R is a future-counterpairing $\to$ R^{-1} is a past-counterpairing
Totality_P	$(\mathsf{P}\phi \wedge \mathsf{P}\psi) \to (\mathsf{P}(\phi \wedge \mathsf{P}\psi) \vee \mathsf{P}(\phi \wedge \psi) \vee \mathsf{P}(\psi \wedge \mathsf{P}\phi))$	If R and S are past-counterpairings, then either $R \circ S^{-1}$ or $S \circ R^{-1}$ is a past-counterpairing, or $R = S$.
Totality_F	$(\mathsf{F}\phi \wedge \mathsf{F}\psi) \to (\mathsf{F}(\phi \wedge \mathsf{F}\psi) \vee \mathsf{F}(\phi \wedge \psi) \vee \mathsf{F}(\psi \wedge \mathsf{F}\phi))$	If R and S are future-counterpairings, then either $R \circ S^{-1}$ or $S \circ R^{-1}$ is a future-counterpairing, or $R = S$.
Density_P	$\mathsf{P}\phi \to \mathsf{PP}\phi$	If R is a past-counterpairing, then $R = S \circ T$ for some past-counterpairings S and T
Density_F	$\mathsf{F}\phi \to \mathsf{FF}\phi$	If R is a future-counterpairing, then $R = S \circ T$ for some future-counterpairings S and T

Table 4.2. Some tense-logical axioms, with corresponding constraints on counterpairings

(ii) Besides the logical relations noted in the discussion of the modal axioms, the tense-logical axioms stand in some further logical relations. Using the rule of necessitation (or 'eternalisation') and the minimal tense logic comprising Mixing_P, Mixing_F, K_P and K_F, the following pairs are interderivable:

4_P and 4_F;	Density_P and Density_F;
NI_P and ND_F;	NI_F and ND_P;
NO_P and NNO_F;	NO_F and NNO_F;
BF_P and CBF_F;[15]	BF_F and CBF_P;
$BF2_P$ and $CBF2_F$;	$BF2_F$ and $CBF2_P$

Corresponding equivalences hold between the constraints in the second column of 4.2. In particular, since COUNTERPAIRINGS automatically yields both directions of CBF and CBF2 without the need for any further constraints, the minimal constraints corresponding to K and Mixing—i.e. that past- and future-counterpairings have universal domains, and are one anothers' converses—suffice to yield the constraints

[15]This is to be expected given the standard Kripkean model-theory, in which BF_P is equivalent to the condition that the domain of each instant is a superset of the domain of each earlier instant, while CBF_F is equivalent to the condition that the domain of each instant is a subset of the domain of each later instant. We can give a direct proof, as follows. Let (*) and (**) be the following schemas:

(*) $$\mathsf{F}\forall x\phi(x) \to \forall x\mathsf{F}\phi(x)$$

(**) $$\exists x\mathsf{G}\phi(x) \to \mathsf{G}\exists x\phi(x)$$

We establish the equivalence of BF_P and CBF_F by way of the following lemmas:

(a): (*) is equivalent to (**). By Duality.

(b): *CBF_F entails (*).* Note that $\forall x(\forall y\phi(y) \to \phi(x))$ is a truth of predicate logic, so we have $\mathsf{G}\forall x(\forall y\phi(y) \to \phi(x))$, and hence by CBF_F, $\forall x\mathsf{G}(\forall y\phi(y) \to \phi(x))$. Thus on the assumption that $\mathsf{F}\forall y\phi(y)$, we can derive that $\forall x\mathsf{F}\phi(x)$ (using K_F), establishing (*).

(c): *(**) entails CBF_F.* Suppose that $\mathsf{G}\forall x\phi(x)$. Then we can introduce a redundant quantifier to get $\forall y\mathsf{G}\forall x\phi(x)$. We can combine this with $\forall y\mathsf{G}(\neg\phi(y) \vee \phi(y))$ to get $\forall y\mathsf{G}(\forall x(\phi(x) \wedge \neg\phi(y)) \vee \phi(y))$, and hence $\forall y\exists z\mathsf{G}(\forall x(\phi(x) \wedge \neg\phi(z)) \vee \phi(y))$. By (**), this yields $\forall y\mathsf{G}\exists z(\forall x(\phi(x) \wedge \neg\phi(z)) \vee \phi(y))$. But $\phi(y)$ follows in predicate logic from $\exists z(\forall x(\phi(x) \wedge \neg\phi(z)) \vee \phi(y))$, so we have $\forall y\mathsf{G}\phi(y)$, which establishes the relevant instance of CBF_F.

(d) *BF_P entails (**).* Suppose $\exists x\mathsf{G}\phi(x)$. Then by Mixing_P, $\mathsf{GP}\exists x\mathsf{G}\phi(x)$, so by BF_P, $\mathsf{G}\exists x\mathsf{PG}\phi(x)$, and hence by Mixing_F, $\mathsf{G}\exists x\phi(x)$.

(e) *(*) entails BF_P* Suppose that $\forall x\mathsf{H}\phi(x)$. Then by Mixing_F, $\mathsf{HF}\forall x\mathsf{H}\phi(x)$. So by (*), $\mathsf{H}\forall x\mathsf{FH}\phi(x)$, and by Mixing_P, $\mathsf{H}\forall x\phi(x)$, which establishes the relevant instance of BF_P.

corresponding to both directions of BF2—i.e. that counterpairings have universal ranges—and hence also the weaker constraints corresponding to both directions of BF.

(iii) By contrast with the modal case, however, in the tense-logical case there are further entailments between the constraints on counterpairings in the second column which do not correspond to any valid derivations among the axiom-schemas in the first column. Namely: given the constraints corresponding to Mixing_{P} and Mixing_{F} (past-counterpairings and future-counterpairings are one anothers' converses), D_{P} ('there is no first instant') and D_{F} ('there is no last instant') become equivalent to one another, and so do $\text{Totality}_{\text{P}}$ ('time does not branch towards the past') and $\text{Totality}_{\text{F}}$ ('time does not branch towards the future'). These further equivalences are surprising from the standpoint of the standard model theory for tense logic. There is no familiar structural property of models such that among models with that property, the two directions of Totality, or the two directions of D, become equivalent to one another.

4.4 The necessity of identity

Of the standard logical schemas considered in the previous section, the necessity of identity deserves extended consideration, since its failure has often been regarded as a distinctive advantage or disadvantage of counterpart theory.

Two main arguments against NI can be found in Lewis's work. The first appeals to the CTQML version of counterpart theory (roughly, our LCT+). Given that analysis, validating NI would require denying that a single object ever has multiple counterparts in one Lewis-world, which is implausible for various reasons (especially because of qualitatively symmetric Lewis-worlds, as discussed in section 4.1). This argument lapses once we take the independently motivated step of replacing LCT+ with an analysis in terms of counterparts of sequences or counterpairings.

The second argument involves putative counterexamples like (11):

(11) The plastic is identical to the dishpan, but might not have been.

According to Lewis (OPW, pp. 252–3), (11) can be used to say something true. But (11) cannot be true in any context if NI holds in every context. At least, this is so if we treat the descriptions 'the plastic' and 'the dishpan' as rigid, or as having wide scope with respect to the modal operator.[16] To avoid distracting questions about scope, we will do better to consider an example with proper names instead of descriptions:

(12) Goliath is identical to Lumpl, but it could have been the case that Goliath was distinct from Lumpl.

(The background story is from Gibbard 1975: molten bronze is poured into a statue-shaped mould, thereby creating a statue, Goliath, and a lump of bronze, Lumpl. Later, the bronze is dissolved in a bath of acid. Thus Goliath and Lumpl are co-located throughout their histories.)

If we accept BMCT as our analysis of singly *de re* modal claims, there is something dialectically inappropriate about using claims like (12) to motivate an account of multiply *de re* modal claims on which the necessity of identity fails. For the natural way to motivate the claim that Goliath could have been distinct from Lumpl is by appeal to claims like (13):

(13) Lumpl could have survived squashing, whereas Goliath could not have survived squashing.

But given BMCT, (13) cannot be true in any single context, if 'Lumpl' and 'Goliath' refer to the same thing. If it is possible to speak truly and literally in uttering (13), doing so must require "changing the context in mid-utterance", i.e. resolving the context-sensitivity of "could have survived squashing" in a non-uniform way. However, for (12) to work as a counterexample to NI, it would have to be true on some *uniform* interpretations. For as I stressed in chapter 1, when one claims a schema to be "logically valid", one is committed only to its instances being true on all uniform resolutions of context-sensitivity. Even paradigmatic logical falsehoods like 'Mary is ready and Mary is not ready', or 'Bruce Wayne is identical to Batman and Bruce Wayne is strong and Batman is not strong', can be used to express truths when we resolve their context-sensitivity non-uniformly; this does not impugn the validity of the non-contradiction schema '$\neg(\phi \wedge \neg\phi)$'. But what principled grounds could there be for insisting that (12) has true uniform interpretations, if one is not prepared to make the same claim about (13)? Once we anchor ourselves in a single context, say by stipulating that by

[16] Several authors have recently observed that it is very hard to get non-rigid readings for incomplete definite descriptions like these: see Rothschild 2007, Hawthorne and Manley MS.

our current standards 'Goliath could not have survived squashing', and hence also 'If Lumpl is identical to Goliath, Lumpl could not have survived squashing', are true, the temptation to assert (12) evaporates.[17]

(Granted, since (12) only a single occurrence of a modal operator, it is not obvious how we could generate a *non*-uniform interpretation of (12) on which it is true. Perhaps, if we think that (12) can be used to communicate a truth, we will have to regard it as some kind of nonliteral discourse, rather than merely non-uniform. A more adequate discussion of the options here would require a deeper foundational account of the concept of uniform interpretation—see Dorr 2014 for some steps towards such an account.)

Thus, I am not persuaded by Lewis's case against NI. Nor do I see that Lewis has done anything to undermine the force of the standard argument *for* NI (Kripke 1971). If x is necessarily identical to x, and y is x, then y must also be necessarily identical to x!

There is no point in having a terminological debate about whether this counts as a genuine instance of Leibniz's Law. Those who deny the validity of the Kripkean argument while accepting the validity of 'a is red; $b = a$; so b is red' will of course want to use the label 'Leibniz's Law' to stand for something that licenses the latter argument but not the former one. It is not technically challenging for a counterpart theorist to do this. But this terminological achievement does nothing at all to undermine the impression that the argument is, in fact, obviously valid. Endorsing a counterpart-theoretic analysis of modal claims does not mean one has to throw away one's prior intuitions about the validity of arguments involving modal operators! Rather, as we have seen, such intuitions are a crucial guide in determining what it takes for the counterpart relation to hold, or for some ordered pairs

[17] Similar points are made by Sider (2009, p. 28) and Russell (2013).

to be a counterpairing.[18]

For many philosophers, the really interesting thing about modal counterpart theory is its supposed ability to block the arguments that push many philosophers towards rich ontological pictures according to which statues are distinct from the lumps of bronze that coincide with them, people are distinct from their bodies, abstract objects of various sorts are distinct from any kind of set-theoretic construct, and so on. Consider a typical such argument:

(14) a. Lumpl could have survived squashing.
b. Goliath could not have survived squashing.
c. Therefore, Goliath is not Lumpl.

How might counterpart theory help rebut this argument? One standard thought goes like this. The expression 'could have survived squashing' is

[18]One further point. There is no problem understanding a sense of 'counterpairing' under which it applies to non-functions, or of sentences ⌜Schmossibly ϕ⌝ stipulated to be analysed according to Counterpairings using such an interpretation of 'counterpairing'. But 'schmossibly', so understood, is semantically anomalous, in that the truth-value of ⌜schmossibly ϕ⌝ depends on merely *orthographic* features of ϕ. In general, we are free to introduce new words by stipulating them to be semantically equivalent to old words, and we expect the new words so introduced to be intersubstitutable *salva veritate* with the old words they were stipulated to be equivalent to, with the exception of quotation-like contexts. For example, I can stipulate that 'Lavid Dewis' is to be semantically equivalent to the name 'David Lewis'. If we use Counterpairings as our analysis of ⌜Schmossibly ϕ⌝, however, we find that 'Schmossibly David Lewis ≠ David Lewis' is analysed as '$\exists R \exists x_1(\mathbf{C}R \wedge R(\text{David Lewis}, x_1) \wedge x_1 \neq x_1)$', which is false, whereas 'Schmossibly Lavid Dewis ≠ David Lewis' is analysed as '$\exists R \exists x_1 \exists x_2(\mathbf{C}R \wedge R(\text{David Lewis}, x_1) \wedge R(\text{Lavid Dewis}, x_2) \wedge x_1 \neq x_2)$', which is true if there is a counterpairing (in the relevant sense) that includes two distinct ordered pairs having Lewis as first element. So 'schmossibly' with those stipulated truth-conditions creates a quotation-like context, in which even expressions that are stipulated to be semantically equivalent cannot be substituted *salva veritate*. Our ordinary word 'possibly', by contrast, surely does not create a quotation-like context: the truth-value of ⌜Possibly ϕ⌝ is sensitive only to semantic features of ϕ.

While requiring counterpairings to be functional is one way to insure that the resulting operator is not quotation-like, it is not the only way. We could also explore a schema like Counterpairings except that the list $t_1, \ldots, t_n$ is taken to stand for the possibly repeating list of *token* singular terms in ϕ, in order of occurrence. On this analysis, '$\Diamond(Ft \wedge \neg Ft)$' is equivalent to '$\exists R \exists x_1 \exists x_2(\mathbf{C}R \wedge Rtx_1 \wedge Rtx_2 \wedge Fx_1 \wedge \neg Fx_2)$', which can be true if R is non-functional. More interestingly, we could, following Kit Fine's suggestion (2007), make the truth-conditions of ⌜Possibly ϕ⌝ sensitive to semantic *relations* between the terms in ϕ that do not depend on their semantic *values*. It may be that whereas 'Lavid Dewis' bears the same semantic relations to 'David Lewis' that 'David Lewis' bears to itself, this is not generally true of pairs of coreferential names, or of pairs of distinct variables.

context-sensitive; each of the many admissible interpretations of 'has a counterpart which survives squashing' corresponds to an admissible interpretation of 'could have survived squashing'. The choice between 'Lumpl' and 'Goliath' can make a difference by affecting how we find it natural to resolve this context-sensitivity. Because of this, each of the argument's premises is true on its most natural, salient, easily accessible interpretations. But there is no single context (admissible interpretation) relative to which both premises are true: replacing 'Lumpl' with the coreferential name 'Goliath' preserves truth within any one context.

Here is an analogy: although Bruce Wayne is Batman, the sentences 'Bruce Wayne is strong' and 'Batman is not strong' are both most naturally interpreted as expressing truths. Wayne/Batman is very strong for a human being, but has no actual superpowers. It is plausible that referring to him as 'Bruce Wayne' makes a reading of 'strong' as 'strong for a human being' more salient, while referring to him as 'Batman' calls to mind the comparison class of other superheroes.

Advocates of *temporal* counterpart theory have embraced analogous responses to arguments against the "extreme monist" position according to which distinct objects never coincide, even temporarily. A sample such argument:

(15) a. No statue in the room has ever been ball-shaped

b. Some statue-shaped lumps of clay in the room have been ball-shaped

c. Therefore, some statue-shaped lumps of clay in the room are not statues

Again, the suggestion (Sider 1996, Sider 2002) is that temporal counterpart theory will help by positing context-sensitivity in 'has been ball-shaped' associated with context-sensitivity in 'past-counterpart'.

The role of counterpart theory in these strategies for defending various surprising identifications is rather limited: the very same moves are available to anyone who accepts that the relevant modal or temporal vocabulary is context-sensitive.[19] In the modal case, that is just about everyone: the claim that modal vocabulary is context-sensitive (in the minimal sense of 'context-sensitive' explained in chapter 1) is extremely banal, and must be recognised

[19] This point is also made by Hawthorne 2007.

by any sensible semantic account.[20] The claim that temporal operators are context-sensitive (in the way that would be relevant to blocking argument (15)) is more controversial. But it is not obvious that temporal counterpart theorists should accept that claim, and on the other hand, one can certainly conceive of views that accept it without embracing a counterpart-theoretic account of the operators.

The pragmatic effect of word-choice on the resolution of context sensitivity is a gentle pressure. It will often be defeated by the countervailing presumption in favour of uniform resolution of context-sensitivity, especially in contexts where we are discussing arguments whose validity turns precisely on such uniform resolution. It would thus be a serious mistake for philosophers attracted to monism to assume that appeals to such pragmatic effects will allow them to systematically mimic the speeches made by "pluralists" who regard arguments like (14) and (15) as sound.

By contrast, some authors, including Lewis himself in Lewis 1971, envisage a kind of theory which achieves a much more extensive and systematic agreement with the "pluralists", by writing the idea that different singular terms invoke different counterpart relations directly and mechanically into the semantics. On one version of this approach, we claim that each singular term in the language is semantically associated with a "sort", in such a way that coreferential terms may be associated with different sorts. 'Goliath' and 'Lumpl', for example, might be associated with the sorts *work of art* and *lump of matter*. Each sort, in turn, picks out a distinctive counterpart relation; the counterpart relations that figure in the analysis of a given modal sentence depend on the sorts associated with its singular terms.[21] Thus 'Lumpl could have survived squashing' and 'Goliath could not have survived squashing' are, respectively, analysed as 'Lumpl has a lump-of-matter-counterpart that survives squashing' and 'Goliath has no work-of-art-counterpart that survives squashing'. Similarly, given a Lewis-style treatment of multiply *de re* sentences, 'Lumpl could have outlasted Goliath' will be analysed as

[20]Some counterpart theorists make the much more controversial claim that the expression 'it is metaphysically possible that' is context-sensitive. But it is not obvious that counterpart theorists have any more or less reason than anyone else for positing context-sensitivity in this particular piece of jargon. In section 5.4, I will argue that given the things philosophers do in introducing the notion of metaphysical possibility, it is not plausible for counterpart theorists or anyone else to regard 'metaphysically possible' as context-sensitive in any way that would help with arguments like (14).

[21]Lewis expresses this formally in an interesting handwritten document 'Counterpart Theory, Mark II' from 1974: thanks to Steffi Lewis for showing me this.

'Lumpl has a lump-of-matter-counterpart that outlasts some work-of-art-counterpart of Goliath that is in the same Lewis-world'.

Here we are clearly no longer dealing with a pragmatic story about the role of word-choices in making certain interpretations of context-sensitivity more salient than others; we have a fully-fledged semantic mechanism, according to which argument (14) is simply invalid.[22] And that, I would say, is a problem, since the argument obviously *is* valid. Indeed, in order to achieve the desired effect of a systematic endorsement of the kinds of intuitive modal claims upon which pluralists base their case for pluralism, the proposal under discussion will need to reject a great deal of standard logic, including syllogisms like (16):

(16) a. No statue could survive squashing.
b. Every lump of clay could survive squashing.
c. Therefore, no statue is a lump of clay.[23]

Those who like this kind of story may regard it as a disadvantage of Counterpairings, compared to Lewis's original counterpart theory, that it provides no item that could be varied depending on the "sorts" associated with the original singular term: the predicate 'counterpairing' occurs only once in the analysis no matter how many terms occurred in the scope of the original modal operator. However, if we really wanted to, we could modify Counterpairings to provide a semantic role for an association between terms and sorts. In outline: instead of taking 'counterpairing' to apply plurally to

[22]What about sentences like 'There is a statue such that it could have survived squashing'? Presumably the proponent of the view will have a story about how the semantic assignment of a kind to a given variable in a sentence is a by the overall role of the variable in that sentence, including the predicates for which it is an argument. ***Cite Ramachandran; Stalnaker; Gibbard; Gupta . . . ***

[23]It is actually controversial whether the notion of identity figures in (16). If we follow Russell in taking the 'is' in (16c) to be the 'is' of identity, we will think of (16) as containing a covert appeal to Leibniz's Law. On the other hand, if we follow authors like Strawson (1950a) and Graff (2001) in taking the 'is' in 'is a lump of clay' to be predicational, we should think of (16c) as having the simple form 'No *F* is *H*; every *G* is *H*; so no *F* is *G*', whose validity has nothing to do with identity. So proponents of sort-relativity who reject (16) are not simply restricting Leibniz's Law.

Some monists who take 'is a lump of clay' to be predicational may simply accept argument (16). Rejecting Leibniz's Law for 'is a lump of clay', they will accept that some statues are *identical* to lumps of clay while denying that it follows from this that any statues *are* lumps of clay. But this takes us beyond anything that could be achieved by means of some modification of counterpart theory, and perhaps beyond the limits of intelligibility altogether.

ordered *pairs* of objects, we can take it to apply to ordered *triples* of the form ⟨sort, object 1, object 2⟩. Then the analysis of 'Possibly $\phi(t_1, \ldots, t_n)$' will be as follows:

$$\exists R \exists v_1 \ldots \exists v_n (\mathbf{C}R \wedge Rs_1 t_1 v_1 \wedge \ldots \wedge Rs_n t_n v_n \wedge \phi(v_1, \ldots, v_n))$$

where $s_1 \ldots s_n$ are the sorts semantically associated with the original terms $t_1 \ldots t_n$, and the new variables $v_1 \ldots v_n$ are required to be associated with the same sorts. The upshot is that a *single* counterpairing R can suffice to witness the truth of 'Lumpl could have survived squashing but Goliath could not have', or of 'Lumpl could have outlasted Goliath'.

For some limited applications of the sort-relativity idea, we could stay with the conception of counterpairings as ordered pairs, simply rewriting Counterpairings in such a way as to incorporate a special treatment for certain special kinds of terms. Such a move could, for example, be endorsed by a proponent of Lewis's identification of qualitative properties with the sets of their instances.[24] The obvious Leibniz's-Law based objection to this identification is that when something is a member of a set, it is necessary that it is a member of that set (at least if it and the set both exist), whereas it often happens that something has a qualitative property but could have failed to have that property (while still existing). The pragmatic, contextualist response is to say that 'Sets have their members essentially' is true in some contexts and false in others; referring to something as a 'set' tends to put us into a context of the first sort, while referring to the same thing as a 'qualitative property' tends to put us into a context of the second sort. A more mechanical response, appealing to a division of singular terms into equivalence classes, could be developed as follows. Each term in the language is semantically marked as either an *ordinary* term or a *qualitative-property* term. When $\phi(t_1, \ldots, t_n, q_1, \ldots, q_m)$ is a sentence with ordinary terms $t_1, \ldots, t_n$ and qualitative-property terms $q_1, \ldots, q_m$, $\ulcorner$'$\Diamond\phi(t_1, \ldots, t_n, q_1, \ldots, q_n)$'$\urcorner$ is analysed as follows:

$$\exists R \exists v_1 \ldots \exists v_n (\mathbf{C}R \wedge \phi(R[t_1], \ldots, R[t_n], q_1, \ldots, q_m))$$

The qualitative-property terms are simply left untouched by the counterpart-

[24]Such an identification seems hopeless for *non*-qualitative properties: if Fred has only sons, the set of sons of Fred is identical to the set of children of Fred, but if we think that Fred could have had a daughter, we cannot say that the property of being a son of Fred is identical to the property of being a child of Fred.

theoretic transformation. We require that counterpairings have universal domains and are functional, so that we can use the function-symbol notation without ambiguity. We also require that counterpairings treat set-membership as rigid: when s is any set and R any counterpairing, $R[s] = \{R[x] : x \in s\}$. To see how this works, suppose 'Set' and 'Prop' both name the entity that is both the set of all philosophers and the property of being a philosopher, but 'Set' is an ordinary term while 'Prop' is a qualitative-property term. 'Possibly Fred has Prop' is analysed as 'For some counterpairing R, R[Fred] has Prop', which is true so long as Fred has a philosopher among his counterparts. By contrast, 'Possibly Fred is in Set' is analysed as 'For some counterpairing R, R[Fred] is in R[Set]'; because counterpairings treat set-membership as rigid, this is true only if Fred in fact belongs to Set.

The point of all this special-purpose semantic machinery is to be able to systematically endorse ordinary claims about the modal or temporal differences between items belonging to different kinds (e.g. statues and lumps, or properties and sets), while still maintaining the identity of items of one kind with items of the other. One compelling objection to this programme is just to insist that the relevant Leibniz's Law inferences *are* valid. Another compelling objection, made forcefully by Fine (2003), is that the sort-relative semantics cannot in fact be set up in such a way as to do justice to the pattern of intuitive modal (or temporal) claims which motivate it. For example, insofar as we are inclined to judge 'Lumpl could have outlasted Goliath' to be true, we are also inclined to judge 'The first thing on Al's list could have outlasted the second thing on Al's list' true in a situation where Al's list reads 'Lumpl, Goliath'. We are still inclined to judge this true even if the speaker was just guessing and had no idea what was actually written on Al's list. It is very hard to imagine a mechanism which could assign different semantic sorts to the terms 'the first thing on Al's list' and 'the second thing on Al's list' in this sentence.

Note that these are objections to the claim that a semantics based on the idea that terms are sort-relative in a way that can lead to failures of Leibniz's Law in modal or temporal environments is correct for *English*. There is no obvious reason why we could not stipulatively introduce a new language with different kinds of terms, which can behave differently under modal operators even when they are coreferential. And if we were doing the semantics for modal and temporal operators in a pre-existing typed language, there might be more to be said for allowing counterpairings to act differently on different types of terms. And this is not a wholly academic question, since semanticists sometimes do resort to honest-to-goodness typed languages in explaining certain natural-language phenomena. For instance, semanticists who posit tacit 'world variables' and 'time variables' normally think of them as belonging to different types from ordinary pronouns.

4.5 Worlds and instants in counterpart theory

We have arrived at what I take to be the optimal forms for counterpart-theoretic analyses of modal and temporal operators. But there is a lot more than this to the task of giving a metaphysical account of modal and temporal discourse: many things we want to say that clearly have a modal or temporal subject matter resist paraphrase in terms of these standard operators. So, what should we try to analyse next? For several reasons, it makes most sense to set our sights on discourse about *possible worlds* and *instants of time*. First, it is a prominent target: explicit talk about entities called 'times', including instants of time, is quite common in ordinary language; and explicit talk about entities called 'possible worlds' is by now an entrenched part of the professional jargon of philosophers and semanticists. Second, explicit quantification over worlds and instants has been employed in many successful metaphysical/semantic analyses of modal and temporal claims of various kinds (e.g. the standard analyses of counterfactuals due to Stalnaker (1968) and Lewis (1973b)); if counterpart theorists can figure out how to extend their theory to include an account of discourse about worlds and instants, they will be free to incorporate these success stories into their view. Third, as we saw in section 2.1, there is much to be said for a theory of mood and tense in natural language that posits, at the level of syntax that serves as the input for semantics, variables ranging over times and worlds, and quantifiers binding these variables, even in very simple sentences; if a theory of this sort is correct, an account of the meaning of sentences containing such variables is prerequisite for any adequate semantics for natural language. Fourth, we will need to see how to integrate talk of worlds and instants into counterpart theory if we want to understand how counterpart theories are relevant to the questions of modal and temporal elitism versus egalitarianism introduced in chapter 1, since these views were introduced as worlds and instants.

The central constraints on the project of fitting talk of instants and worlds into counterpart theory come from the "roles" for instants and worlds, which section 1.3 attempted to lay out systematically as a collection of schemas. If we could find an interpretation of the relevant theoretical vocabulary ('instant' and 'world'; 'possible', 'future' and 'past'; 'accessible from', 'before' and 'after'; 'at') under which all of the relevant schemas came out as valid, it is hard to see how we could reasonably reject these interpretations in favour of some other interpretations according to which the schemas in question are *not* valid. The schemas are much too central to our use of 'instant' and 'possible world' to be brushed aside in this way. Moreover, many applications of instants and worlds in semantical and metaphysical analyses implicitly rely on these schemata; even if we for some reason thought that the *words* 'possible world' and 'instant' should be understood in a way that violated the schemata, the analyses in question would then need to be redone, perhaps by replacing the vocabulary of worlds and instants with some other vocabulary which did satisfy the roles. Also, the schemas were part of the background which made temporal and modal elitism and egalitarianism look like reasonable ways of capturing the weighty metaphysical issue that is at stake in debates between A- and B-theorists; if we ended up giving some eccentric, role-violating interpretation of the words 'instant' and 'world', that would tend to suggest that our formulations of elitism and egalitarianism were not, after all, good ways to capture the underlying metaphysical dispute.

The natural place to start is with Lewis's proposal that the possible worlds are the Lewis-worlds. By making this identification, Lewis suggests that he can accommodate at least the following central elements of the possible world role:

($\Diamond$) Possibly ϕ iff ϕ at some possible world.

($\Box$) Necessarily ϕ iff ϕ at every possible world.

To evaluate this claim, we will need to know how Lewis proposes to interpret 'At w, ϕ' (or as he sometimes likes to put it, 'According to w, ϕ', or 'w represents that ϕ'). His idea is that when ϕ is *de dicto* (and all its predicates are well-behaved), 'At w, ϕ' is equivalent to $[\phi]^w$, the result of adding a restrictor 'in w' to every quantifier in ϕ; when ϕ involves expressions which refer *de re* to certain particular objects, the analysis of 'At w, ϕ' will also involve quantification over counterparts of those objects in w. Clearly this strategy for making sense of 'at w' has no hope of vindicating $\Diamond$ and $\Box$ unless we adopt the Lewis-style analysis of 'possibly' and 'necessarily' (in which 'possibly' and 'necessarily' introduce quantifiers over Lewis-worlds) which I have already argued against at length, in chapter 3 and in section 4.1.

But even if we set all the problems with this analysis aside, it is highly misleading of Lewis to suggest that it can be seen as the result of plugging a certain interpretation of 'possible world' and 'at' into ◇ and □. For even *given* Lewis's analyses of possibility and necessity, getting ◇ and □ to be true would require interpreting 'at' in two quite different ways: in ◇, 'at *w*' would need to be understood as introducing existential quantification over counterparts, while in □ it would need to be understood as introducing universal quantification. We could afford to ignore this difference only if we thought that each thing had exactly one counterpart in each Lewis-world, a claim that is extremely implausible given Lewis's other commitments.

As we have seen, the best counterpart theoretic analysis of modal operators drops Lewis's ideas about the role of Lewis-worlds, and with it any prospect of being able to think of Lewis-worlds as the occupants of the possible world role. But this doesn't mean we have to give up on ◇ and □! For as we have seen, provided that we require all counterpairings to have *universal domains* and to be *functional*, Counterpairings can be expressed in the following compact form:

$$\Diamond\phi(t_1,\ldots,t_n) \coloneqq \exists R(\mathbf{C}R \wedge \phi(R[t_1],\ldots,R[t_n]))$$
$$\Box\phi(t_1,\ldots,t_n) \coloneqq \forall R(\mathbf{C}R \rightarrow \phi(R[t_1],\ldots,R[t_n]))$$

This looks to be of the right logical form to be a specification of ◇ and □. We need only understand 'possible world' to mean 'counterpairing', and adopt the following analysis of what it is for something to be the case "at *R*", where *R* denotes a plurality of ordered pairs which is functional and has a universal domain:

(At *R*) $$\text{at}\, R\, \phi(t_1,\ldots,t_n) \coloneqq \phi(R[t_1],\ldots,R[t_n])$$

The situation is analogous in the temporal case. The natural first thought is that the "instant" role will be played by certain three-dimensional regions of spacetime ('Slices'). But this turns out to be deeply problematic. Even if we only cared about the most basic part of the instant role, as given by principles like At-F and At-P

(At-F) It will be that ϕ iff at some future instant, ϕ

(At-P) It has been that ϕ iff at some past instant, ϕ.

the proposal could work only if we adopt a version of temporal counterpart theory modelled closely on Lewis's counterpart theory, with Slices standing in for Lewis-worlds. But this is a bad idea for the reasons given in chapter 3 and in the first section of this chapter. And even waiving all those reasons, the only way to make things work with a univocal interpretation of 'at' would require the problematic assumption that each thing has exactly one counterpart 'in' each Slice. A better idea falls right out of the form of Counterpairings: require future-counterpairings and past-counterpairings to be functional and universally defined; analyse 'at' using (At R); and identify future and past instants with future-counterpairings and past-counterpairings respectively.

There is a complication: officially, counterpairings are not entities. That is, 'counterpairing' is officially a plural predicate; quantification over counterpairings is plural quantification. So claims like 'Every possible world is a counterpairing' and 'Every future instant is a future-counterpairing' are not even grammatically well-formed. Whether we have to worry about this depends on our purposes. For many of the applications of worlds and instants, it would be enough to be able to analyse the sentences in a *typed* language with special syntactic categories of 'instant variables' or 'world variables' alongside ordinary variables. And for this project, it doesn't matter at all if our analysis involves translating these special variables into plural variables rather than singular ones. This kind of "reconstructive analysis" would, for example, suffice for the purpose of integrating some Stalnaker- or Lewis-style analysis of counterfactual conditionals with a counterpart theoretic account of 'necessarily' and 'possibly': nothing is lost from these analyses if we replace the quantifiers over possible worlds with plural quantifiers over counterpairings and take 'closer than' to be a predicate taking plural arguments. A reconstructive analysis would also suffice for the purpose of providing a counterpart-theoretic semantics for natural languages on the assumption that they contain syntactically real 'time variables' and 'world variables' (see section 2.1). On the other hand, there are other purposes for which we would require a "direct reduction", which lets us identify certain particular entities as the worlds or instants. A direct reduction would be needed for an analysis of English discourse involving the words 'instant' and 'possible world', which are obviously in the same grammatical category as 'dog' and 'house'. It would be also be required for applications that require us to speak of *sets* of worlds and instants, *functions* defined on all worlds and instants, etc.

Even if we decide we want a direct reduction as opposed to a reconstruction, there is a natural way to hold on to the basic idea behind the identification of worlds or instants with counterpairings. Suppose that we find some domain of entities each of which corresponds, in some natural way, to some ordered pairs which are functional and have universal domain. And suppose that whenever some ordered pairs are a counterpairing, they correspond in the relevant way to one of these entities. Then we could think of these entities as being the worlds [instants], with the *possible* worlds, [*future* and *past* instants] being those corresponding to pluralities of ordered pairs which are [future- and past-] counterpairings. Let $P(x)$ be a plural term that stands for whichever plurality of ordered pairs which "correspond" to the relevant entity x. Then we can analyse 'at x, ϕ' as 'at $P(x)$, ϕ' in the sense of (At R), and thereby recover $\Diamond$ and $\Box$, or F and P.

What sorts of entities might do this job of "corresponding" to counterpairings? The most obvious candidates are *sets* of ordered pairs—*functions*, as standardly conceived. Suppose there is a set U of all non-sets, and let f be some function whose domain is U. Then given that sets are well-founded, there is a natural way to extend the action of f to *any* set. We simply take the image of a set under f to be the set of the images of its members under f. That is to say, f corresponds in a natural way to a plurality of ordered pairs $P(f)$ that has a universal domain and is functional, where the correspondence is defined as follows:

$$\begin{aligned}&P(f)x_1x_2 :=\\ &\quad \exists g(\text{Function}(g) \wedge f \subset g \wedge x_2 = g(x_1) \wedge \forall y(\text{Set}(y) \to g(y) = \{g(z)|z \in y\}))\end{aligned}$$

Of course, not all pluralities of ordered pairs can be generated in this way from sets of ordered pairs. But consider the plausible assumption that *sets have their members essentially*:

$$\forall x \forall y((x \in y \to \Box x \in y) \wedge (x \notin y \to \Box x \notin y))$$

Given Counterpairings, this is equivalent to the following principle about counterpairings:

Rigid Sets Whenever R is a counterpairing and s is a set, $R(s) = \{R(x)|x \in s\}$

If Rigid Sets is true, then the *counterpairings* are among those pluralities of ordered pairs that *can* be generated from sets of ordered pairs. So we could take the *sets* to be the worlds [instants], understanding 'x is a possible world [future instant/past instant]' as equivalent to '$P(x)$ is a counterpairing [future-counterpairing/past-counterpairing]', and understanding 'at f ϕ' as equivalent to 'at $P(f)$ ϕ' in the sense of (AT R).

This approach works on the controversial assumption that there is a set of all non-sets. But even if this assumption is false, we may be able to use a generalisation of this approach in defining what it is for a plurality of ordered pairs to be "generated" by a set. For it may be that among all the things there are, we can find some set-sized minority of 'basic' entities, such that everything else is 'built up' from them by means of relations analogous to set-membership in being modally and temporally rigid, and well-founded. If so, we can still take the occupants of the world or instant role to be sets of ordered pairs, and define '$R = P(f)$' by means of a disjunctive definition with one clause for the "basic" entities and another for each "building" relation.

If there were nothing more to the world role than $\Diamond$ and $\Box$, and no more to the instant role than F and P, we would be finished. But as we saw in section 1.3, there is quite a bit more involved. First, there are principles about the interaction of 'at' with truth-functional operators:

(At-$\wedge$) At i: (ϕ and ψ) iff at i: ϕ and at i: ψ

(At-$\vee$) At i: (ϕ or ψ) iff at i: ϕ or at i: ψ

(At-$\neg$) At i: not-ϕ iff not (at i: ϕ)

So long as the pluralities corresponding to worlds and instants always have universal domains and are always functional, each of these principles is automatically satisfied given (AT R).[25]

[25]What about the following more controversial principles?

(At-=) At i, $x = y$ iff $x = y$

(At-$\forall$) At i, everything ϕs iff everything is such that at i, it ϕs

(At-$\exists$) At i, something ϕs iff something is such that at i, it ϕs

Given that the pluralities corresponding to worlds and instants are functional, the right-to-left direction of At-= is automatically satisfied; and given that they have universal domains, the left-to-right direction of At-$\forall$ and the right-to-left direction of At-$\exists$ are also satisfied. (This is related to the fact that CBF is a consequence of COUNTERPAIRINGS.) The other direction of At-= would follow if we made the further postulate that the relevant pluralities are always *one-to-one*, and the other directions of At-$\forall$ and At-$\exists$ would follow if we made the further postulate that the relevant pluralities always have universal *ranges*. In the temporal case, these further postulates are guaranteed by the fact that future- and past-counterpairings are one anothers' inverses, given that they are functional and have universal domains.

Another important aspect of the world and instant roles concerns the *accessibility* and *beforeness* relations. In the case of worlds, this is given as follows:

(At-$\Diamond$) At w: possibly ϕ iff ϕ at some w accessible from w

(At-$\Box$) At w: necessarily ϕ iff ϕ at every w' accessible from w

The challenge posed by these principles is to find a meaning for 'accessible from' that makes it true. We can find one by substituting our analysis of 'possibly' or 'necessarily' (Counterpairings) into our analysis of 'at' (At R). In the simple case where 'counterpairing' is interpreted as qualitative, this is easily done:

$$\begin{aligned}
&\mathsf{at}\,R\,\Diamond\phi(t_1,\ldots,t_n)\\
&\leftrightarrow\ \mathsf{at}\,R\,\exists Q(\mathbf{C}Q\wedge\phi(Q[t_1],\ldots,Q[t_n]))\\
&\leftrightarrow\ \exists Q(\mathbf{C}Q\wedge\phi(Q[R[t_1]],\ldots,Q[R[t_n]]))\\
&\leftrightarrow\ \exists Q(\mathbf{C}Q\wedge\mathsf{at}(Q\circ R)\,\phi(t_1,\ldots,t_n))
\end{aligned}$$

Thus our reconstruction of worlds as pluralities of ordered pairs can accommodate At-$\Diamond$ (and by duality, At-$\Box$) provided we adopt the following analysis of 'S is accessible from R':

$$(<)\qquad R < S := \exists Q(\mathbf{C}Q \wedge S = Q\circ R)$$

If we are giving a direct reduction instead of a reconstruction, we can say that $x < y$ iff $P(x) < P(y)$ in the sense of (<). Note that if we also require the pluralities corresponding to worlds to be one-to-one, the right-hand side of (<) is equivalent to '$\mathbf{C}(S\circ R^{-1})$'. In the temporal case, we know that the pluralities of ordered pairs corresponding to future and past instants always *are* one to one, since this follows from functionality together with the requirement that future- and past-counterpairings are one anothers' inverses. So we can understand 'R is before S' to mean '$S\circ R^{-1}$ is a future-counterpairing', or equivalently, '$R\circ S^{-1}$ is a past-counterpairing'.[26]

[26]What about non-qualitative interpretations of 'counterpairing', e.g. those associated with doxastic possibility for a particular person? Suppose $\mathbf{C}S$ is analysed as $\psi(a_1,\ldots,a_j,S)$, where ψ is qualitative and $a_1\ldots a_j$ are some singular or plural terms. Then, in the third and fourth lines of the above derivation, instead of '$\mathbf{C}Q$' we should have '$\psi(R[a_1],\ldots,R[a_j],Q)$'; so our analysis of '$S$ is accessible from R' should be '$\exists Q(\psi(R[a_1],\ldots,R[a_j],Q)\wedge S=Q\circ R)$'. If

So far, then, things are going as well as we could wish for the proposal to identify worlds and instants with pluralities of ordered pairs, or with entities which correspond in some natural way to such pluralities. But trouble arises when we turn to the final component of the roles from section 1.3, the claim that facts about what is the case at a world [instant] are necessary [eternal]:

(Invariance) If at w, ϕ then necessarily, at w, ϕ

(Permanence) If at t, ϕ then always, at t, ϕ

As we saw in chapter 1, many of the standard ways of talking about worlds and instants presuppose Invariance and Permanence, and these principles need to be in the background if questions T2 and M2 are to have their envisaged significance in articulating the metaphysical issue in about which A-theorists and B-theorists disagree.[27] It is unfortunate, then, that Invariance and Permanence turn out to be *false* when 'at' is analysed in accordance with (At R), given the very plausible assumption that *ordered pairs are rigid*:

$$\forall x \forall y \forall z(z = \langle x, y\rangle \rightarrow \Box z = \langle x, y\rangle)$$

Given Counterpairings, this is equivalent to the following principle about counterpairings:

Rigid Pairs Whenever R is a counterpairing, $R[\langle x, y\rangle] = \langle R[x], R[y]\rangle$.

It follows from Rigid Pairs that a counterpairing must map each identity pair to another identity pair. Hence each counterpairing must map I, the

counterpairings are global permutations, then we can express this analysis in terms of $\mathbf{C}$: in that case, '$\exists Q(\psi(R[a_1], \ldots, R[a_j], Q) \wedge S = Q \circ R)$' is equivalent to '$\psi(R[a_1], \ldots, R[a_j], S \circ R^{-1})$', which is equivalent to '$\psi(R[a_1], \ldots, R[a_j], R[R^{-1}[S \circ R^{-1}]])$', which is equivalent to 'at R $\mathbf{C}(R^{-1}(S \circ R^{-1}))$'.

[27] However, not all of the theoretical applications for worlds and instants require Invariance/Permanence. For example, even if facts about what is the case at a plurality of ordered pairs are contingent, we could still invoke quantification over such pluralities in a Stalnaker- or Lewis-style account of 'If it were that ϕ it would be that ψ' as 'Either there is no counterpairing at which ϕ, or some counterpairing at which $\phi \wedge \psi$ is closer than any counterpairing at which $\phi \wedge \neg\psi$'. (Since we are analysing 'If it were that ϕ, it would be that ψ', rather than 'At w: (if it were that ϕ, it would be that ψ)', we use a two-place closeness relation.) If we were thinking of the relata as possible worlds (subject to Invariance), we would have to think of the extension of this relation as a contingent matter: at each world w, w is the closest world (or at least, among the closest worlds). However, if we give up on Invariance, we can think of the extension of 'closer than' as a necessary matter: the more similar a counterpairing is to the identity (in contextually relevant respects), the "closer" it is.

plurality of all identity pairs, to some plurality J of identity pairs. (If counterpairings have to have universal ranges, $J = I$.) But the conjunction of 'at $J\,\phi$' with the claim that J consists only of identity pairs entails ϕ. Thus for any counterpairing R, if 'at R at $I\,\phi$' is true, then 'at $R\,\phi$' is true. So it is necessarily [eternally] true that if at $I\,\phi$, then ϕ. (If counterpairings have universal ranges, the converse also holds). Thus whenever it is contingently [temporarily] true that ϕ, it must also be contingently [temporarily] true that at $I\,\phi$.

Given Rigid Pairs, our reconstructive analysis of quantification over worlds or instants as plural quantification over ordered pairs runs aground on Invariance and Permanence. What about a direct reduction, in which worlds or instants are identified with entities which somehow 'correspond' to counterpairings? Clearly, given Rigid Pairs, this will allow us to accept Invariance and Permanence only if the "correspondence" relation which takes us from one of the relevant entities x to a plurality of ordered pairs $P(x)$ is a contingent [transient] one. For example, consider the entity i such that $P(i) = I$. If it were necessary that $P(i) = I$, the necessity of $\ulcorner \text{at}\, I\,\phi \to \phi \urcorner$ would yield the necessity of $\ulcorner \text{at}\, i\,\phi \to \phi \urcorner$, and hence the contingency of $\ulcorner \text{at}\, i\,\phi \urcorner$ for contingent ϕ. To preserve the necessity of at $i\,\phi$, we need it to be contingent that $P(i) = I$. Assuming that *sets* are rigid, this rules out our earlier suggestion that the entities should be taken to be sets of ordered pairs. And it is not at all clear what else they could be.

Giving up Rigid Pairs would be a heavy cost. But if we were prepared to pay this cost, could we get Invariance/Permanence to come out true when 'At w'/'At t' are replaced with 'At R', analysed in accordance with (At R)? Yes: it turns out that everything works out nicely if we (a) require all counterpairings not only to be functional and have universal domain, but also to be one-to-one and have universal range—i.e. to be *global permutations*, and (b) reject Rigid Pairs in favour of the following rather bizarre alternative:

Semi-Rigid Pairs Whenever R is a counterpairing, $R[\langle x, y\rangle] = \langle R[x], y\rangle$.

To see how this works out, note first that Semi-Rigid Pairs entails that whenever R is a counterpairing and S is any plurality of ordered pairs, $\langle x, y\rangle$ is one of $R[S]$ iff $x = R[z]$, for some z such that $\langle z, y\rangle$ is one of S. In other words, $\forall x \forall y R[S]xy \leftrightarrow \exists z(Rzx \wedge Szy)$. More succinctly still,

$$R[S] = S \circ R^{-1} \tag{17}$$

Now, consider what it takes for 'at R at $S\,\phi(t_1, \ldots, t_n)$' to be true under these assumptions, when R is a counterpairing and S is any plurality of ordered pairs that is functional and has a universal domain. According to (At R), 'at R at $S\,\phi(t_1, \ldots, t_n)$' is equivalent to 'at $R\,\phi(S[t_1], \ldots, S[t_n])$', and hence to

$$\phi(R[S][R[t_1]], \ldots, R[S][R[t_n]]).\text{[28]} \tag{18}$$

Since R is a counterpairing, (17) tells us that $R[S] = S \circ R^{-1}$. Hence $R[S][R[t_i]] = (S \circ R^{-1})[R[t_i]] = S[R^{-1}[R[t_i]]]$. But given our assumption that counterpairings are one-to-one, $R^{-1}[R[t_i]] = t_i$. (Note that this reasoning works whether t_i is a singular or plural term.) So (18) is equivalent to '$\phi(S[t_1], \ldots, S[t_n])$', and hence to 'at $S\,\phi(t_1, \ldots, t_n)$'. We have thus shown that if 'at $S\,\phi(t_1, \ldots, t_n)$' is true, 'at R at $S\,\phi(t_1, \ldots, t_n)$' is true for every counterpairing R, so that 'at $S\,\phi(t_1, \ldots, t_n)$' is necessarily [always] true.

To elucidate this result, let us see how Semi-Rigid Pairs secures the non-contingency of 'at $R\,\phi$' in a particular case. Consider the claim that at I, Lewis is a philosopher (where I is the plurality of all identity pairs). Since it is is contingent whether Lewis is a philosopher, and since all I does is map Lewis to himself, how can it be necessary that at I, Lewis is a philosopher? The answer is that, given Semi-Rigid Pairs, the ordered pairs in I are not *necessarily* identity pairs. For example, the identity pair ⟨Lewis, Lewis⟩ is not necessarily an identity-pair, since whenever x is a counterpart of Lewis, ⟨x, Lewis⟩ is a counterpart of ⟨Lewis, Lewis⟩. But since all its counterparts have Lewis himself (not just one of his counterparts!) as their second member, all its counterparts have a philosopher as their second member: thus ⟨Lewis, Lewis⟩ is *necessarily* a pair whose second member is a philosopher. Likewise, it is not necessary that I are all identity pairs, but it is necessary that whichever of I has Lewis as a first member has a philosopher as its second member. Indeed, it would have been *very easy* for I not to have been identity pairs—some of I would have failed to be identity pairs if *anything* had been otherwise in any respect. For whenever it is true that ϕ, it is necessary that at I, ϕ; and it is always necessary that (ϕ iff at whichever pairs are the identity pairs, ϕ); hence, whenever it is true that ϕ, it is necessary that (if not-ϕ, then I are not the identity pairs). *Every* counterpairing consists of identity pairs *at itself*: by Rigid Pairs, $R[R] = R \circ R^{-1} = I$ (since counterpairings have universal domains and are bijective). Thus 'comprises all and only the identity pairs' plays the role, in this theory, of the predicate 'is actualised'

[28]Note that we get the same result if we first eliminate 'at R' to get 'at $R[S]\,\phi(R[t_1], \ldots, R[t_n])$' and then eliminate 'at $R[S]$'.

in the theory of possible worlds. This takes some getting used to: one must constantly fight the temptation to slip back into the picture of ordered pairs as rigid, which is deeply entrenched in our ordinary ways of thinking about ordered pairs.

Our problem, then, is that if we want to identify possible worlds or instants with pluralities or sets of ordered pairs, and retain Invariance or Permanence, we have to give up the extremely plausible view that ordered pairs are rigid in favour of the outlandish Semi-Rigid Pairs. There are several possible responses to this problem that do not require us to embrace Semi-Rigid Pairs. One response is that it just goes to show that we should not identify possible worlds with those particular pair-theoretic or set-theoretic constructs. After all, theories which identify objects of this or that kind (properties, propositions, events, processes, aspects, absences. . .) with set-theoretic constructs are not all that popular these days. And one common line of objection to many such theories involves the charge that the target entities have different modal properties from the set-theoretic constructs to which they are supposed to be identical. For example, this looks like a powerful objection to Lewis's identification of properties with sets, even when this is restricted to qualitative properties. Those who committed in any case to regarding the realm of abstract entities as much more extensive than the realm of sets will naturally react to the problem by keeping Rigid Pairs and rejecting the identification of possible worlds with sets or pluralities of pairs.

What *are* the entities that play the role of worlds or instants, if they are not sets of ordered pairs? If we are generally hostile to reductionism in ontology, there may not be much more to say: we describe the characteristic role played by the entities in question, which includes a certain modal profile, and that's that. However, it might be helpful to consider a version of the view that departs minimally from the proposals using ordered-pairs. Suppose that we expand our ontology to include several different kinds of ordered-pair-like objects, all governed by the basic structural principle that for any objects x and y, there is exactly one such object whose 'first element' is x and whose 'second element' is y, but having different modal profiles. As well as ordered pairs, there are ordered pairs*; for each ordered pair $\langle x, y\rangle$, there is a corresponding ordered pair $\langle x, y\rangle^*$. Semi-Rigid Pairs is false, but the corresponding principle about pairs* is true. We can then identify worlds or instants with pluralities or sets of pairs* rather than pairs. The natural correspondence between pairs and pairs* generates a natural correspondence between such sets/pluralities and the pluralities of *pairs* to which our predicate 'counterpairing' continues to apply.

Of course lovers of set-theoretic reductionism, like Lewis, will not be satisfied with this approach. A more Lewisian strategy would be to keep the identification of possible worlds with sets or pluralities of ordered pairs, while deploying Lewis's idea that choices between different ways of referring to the same thing(s) can affect the truth value of our modal claims. Here is a set *s* of ordered pairs, which is also a possible world. If we refer to *s* as "a set of ordered pairs", that will tend to put us into a context in which Rigid Pairs is true. If we refer to *s* as "a possible world", that will tend to put us into a different context, relative to which Semi-Rigid Pairs will be true, so that claims about what is the case "at" *s* become non-contingent. If we try to do both these things in the same breath, we will be misled into thinking that we are thinking with two distinct entities.

As we saw in section 4.4, the ability of such appeals to context-sensitivity to vindicate our ordinary modal discourse is limited. It is fairly common in philosophy to find speeches which simultaneously presuppose ordered-pair-theoretic essentialism and the idea that facts about what is the case at a world are non-contingent. For example, people will say things like 'At every possible world, the elements of an ordered pair are the same elements it has at the actual world'. The only way for the present approach to save such speeches would require appealing somehow to non-uniform resolution of context-sensitivity; and this will be quite a stretch, since the claims in question don't have the distinctive feel of discourse that involves "mid-sentence context-switching".

This is a problem for the use of the context-sensitivity idea to sustain a direct reduction of possible worlds or instants to sets of ordered pairs. It is not, however, a problem for the use of the idea within a reconstructive analysis, in which the aim is to translate a many-sorted language with a syntactically distinct category of world variables (or instant variables). For as we saw in section 4.4, once we are dealing with languages whose terms and variables come in different kinds, it is easy to modify or extend Counterpairings in such a way that terms of different kinds get treated differently when they occur in the scope of modal/temporal operators. (Of course, for any such modification of Counterpairings, we will need a corresponding modification of (At R) to preserve principles like At-$\Diamond$ and At-$\Box$). In particular, we can modify Counterpairings and (At R) in such a way that ordinary terms and variables get treated as before, while world/instant terms and variables are treated in a way that automatically achieves the effect of Semi-Rigid Pairs.

Let us sketch what it would look like to work this strategy out rigorously. We start with a language $\mathcal{L}_W$ with three sorts of terms: ordinary singular terms, ordinary plural terms, and world variables. Besides quantifiers, truth-functional connectives, predicates taking ordinary terms as arguments (including a one-place plural predicate **C**, 'is a counterpairing') and the ordered-pair-forming functor, $\mathcal{L}_W$ contains an 'at' operator that takes a world variable and a sentence and delivers a sentence, and two predicates P ('is a possible world') and A ('is actualised'), each of which takes one world variable as its argument. We want to translate each sentence of $\mathcal{L}_W$ into a sentence of $\mathcal{L}$, the sublanguage of $\mathcal{L}_W$ with only ordinary singular and plural terms. The translations will proceed via an intermediary language $\mathcal{L}^+$, which is like $\mathcal{L}_W$ except that world terms are allowed to occur in any syntactic position where a plural term could occur. We proceed in two stages. At the first stage, we replace each formula ϕ of $\mathcal{L}_W$ with a formula ϕ' of $\mathcal{L}^+$ in which the 'At' operator does not occur; at the second stage, we will eliminate all the world variables from ϕ' leaving us with a formula ϕ'' of $\mathcal{L}$.

To eliminate the 'At' operator at stage one, we will apply the following analysis as many times as necessary:

(19)

$$\text{at}\, W_0\, \phi(t_1, \ldots, t_n, W_1, \ldots, W_k) := \phi(W_0[t_1], \ldots, W_0[t_n], W_1 \circ W_0{}^{-1}, \ldots W_k \circ W_0{}^{-1})$$

Here W_0 is any world variable, and $\phi(t_1, \ldots, t_n, W_1, \ldots, W_k)$ stands for any formula of $\mathcal{L}^+$ with ordinary terms (singular and plural) $t_1, \ldots, t_n$ and world variables $W_1, \ldots, W_k$, in order of first occurrence. By repeatedly applying (19), we can turn each sentence of $\mathcal{L}_W$ into an 'at'-free sentence.

(As usual, we treat sentences containing complex terms like '$R[t_i]$' and '$W_i \circ W_0{}^{-1}$' in (broadly) Russellian fashion, as abbreviations of quantified sentences. So, for example, the result of applying (19) to at $W_0\, \phi(x_0, W_1)$, i.e. $\phi(W_0[x_0], W_1 \circ W_0{}^{-1})$, can be taken to be an abbreviation for the following formula of $\mathcal{L}^+$:

$$\begin{aligned}&\exists x_1 \exists W_2 (W_0\langle x, y\rangle \wedge \\ &\qquad \forall x_2 \forall x_3 (W_2\langle x_2, x_3\rangle \leftrightarrow \exists x_4 (W_0\langle x_4 x_2\rangle \wedge W_1\langle x_4 x_3\rangle))) \wedge \phi(x_1, W_2))\end{aligned}$$

Note that this in effect gives each complex term the widest scope it could possibly have.)

At stage two, we want to transform the 'At'-free formula ϕ' of $\mathcal{L}^+$ into a formula ϕ'' of $\mathcal{L}_{\mathrm{ACT}}$. We do so by uniformly replacing each world variable W with a plural variable V that does not already occur in ϕ', while restricting the quantifier binding V to pluralities of ordered pairs which are global permutations. That is, $\exists W_i(\ldots)$ will be replaced by $\exists V(\forall x \exists y V\langle x, y\rangle \wedge \forall z(V\langle x, z\rangle \rightarrow z = y) \wedge \exists y(V\langle y, x\rangle \wedge \forall z(V\langle z, x\rangle \rightarrow z = y)) \wedge \ldots)$, where x, y, and z are singular variables that do not already occur in ϕ'. Finally, we need to eliminate occurrences of the predicates P ('is a possible world') and A ('is actualised'), which take world arguments in $\mathcal{L}_W$ and hence do not occur in $\mathcal{L}$. We do this by replacing each formula 'PV' with '$\mathbf{C}V$' ('V is a counterpairing'), and replacing each formula of the form 'AV' with '$\forall x(Vx \leftrightarrow \exists y(x = \langle y, y\rangle))$' (where x and y are singular variables that do not already occur in ϕ_1).

Here is an example of the machinery at work. Suppose our original sentence ϕ of $\mathcal{L}_{\mathrm{ACT}}$ is

(20) $$\exists W_0(AW_0 \wedge \exists W_1(PW_1 \wedge \text{at } W_1\ (\exists x(F(x) \wedge \neg \text{ at } W_0\ F(x)))))$$

We can render this in English as 'there is a possible world at which there is an F thing that is not F at the actual world'. Eliminating the two occurrences of 'At' in reverse order yields first

(21) $$\exists W_0(AW_0 \wedge \exists W_1(PW_1 \wedge \text{at } W_1\ (\exists x(F(x) \wedge \neg F(W_0[x])))))$$

and then

(22) $$\exists W_0(AW_0 \wedge \exists W_1(PW_1 \wedge \exists x(F(x) \wedge \neg F(W_0 \circ {W_1}^{-1})[x])))$$

(Note that we would get a logically equivalent result if we eliminated the occurrences in the opposite order.) That concludes step one. At step two, we convert this (abbreviation of a) formula of $\mathcal{L}^+$ into the following (abbreviation of a) formula of $\mathcal{L}$:

(23) $$\exists X_0(\mathrm{Perm}(X_0) \wedge \forall x'(X_0 x' \leftrightarrow \exists y x' = \langle y, y\rangle) \\ \wedge \exists X_1(\mathrm{Perm}(X_1) \wedge \mathbf{C}X_1 \wedge \exists x(F(x) \wedge \neg F(X_0 \circ {X_1}^{-1})[x])))$$

(Here 'Perm(X)' means that X is a global permutation.) Given standard plural logic and the theory of ordered pairs, this formula is equivalent to

(24) $$\exists X_1(\mathrm{Perm}(X_1) \wedge \mathbf{C}X_1 \wedge \exists x(F(x) \wedge \neg F(({X_1}^{-1}[x]))))$$

and hence to

$$\exists y \exists X_1(\neg F(y) \wedge \mathrm{Perm}(X_1) \wedge \mathbf{C}X_1 \wedge F(X_1[y])) \tag{25}$$

which is what we would have expected.

4.6 Actuality

Consider Twin, off in some other Lewis-world. He is about six feet tall. He could have been much shorter than that, say five feet tall. And if so, he would have been much shorter than he in fact is, since in fact he is about six feet tall.

'Actually' and 'in fact' are roughly synonymous. The way Twin in fact is is the way he actually is. Once we agree that Twin is about six feet tall, it would be absurd to deny that he is *in fact* about six feet tall. It would be no less absurd to deny that he is *actually* six feet tall.

When Lewis discusses actuality, he generally focuses on the adjective 'actual' rather than the adverb 'actually'. He claims that only things that are in Cosmo are actual; things that are in other Lewis-worlds are non-actual. 'Actual', according to him, is indexical. As used by us, its extension includes only things that are parts of Cosmo; as used by someone in another Lewis-world, its extension includes only things that in that world.

I am not quite sure how 'actual' is supposed to relate to 'actually'. I am not even sure that 'is actual', as opposed to 'is an actual *F*', is part of standard English. But it would certainly be strange to agree with Lewis that Twin is not actual, while accepting (as I claimed we must) that he is actually six feet tall. Maybe we should say to be actual is to be actually identical to something. If so, then for the same reason that I think we must say that Twin, being six feet tall, is actually six feet tall, I think we must say that Twin, being identical to something, is actually identical to something, and thus actual.

If we thought that Lewis-worlds were *ways things might be*, we would naturally be led to think that there must be one Lewis-world that is *the way things are*, and thus to give this Lewis-world a special role in the semantics of 'actual' and 'actually'.[29] But once we have absorbed the lesson that the

[29]Lewis says that he 'would find it very peculiar to say that modality, as ordinarily understood, is quantification over parts of actuality. If I were convinced that I ought to call all the worlds actual. . . then it would become very implausible to say that what might happen is what does happen at some or another world' (*OPW* 100). It is not obvious how to square this with his allowing (*OPW* 231–2) that actual objects can be one another's counterparts.

concept *Lewis-world* plays no special role in the semantics for modal claims, we will see that this is a mistake. Lewis-worlds, like everything else, are counterparts, and in that sense represent ways certain specific things—the things whose counterparts they are—might be. If the Lewis-worlds are all one another's counterparts, then each Lewis-world represents a way any other given Lewis-world might be.[30] There is only one Lewis-world—namely Cosmo itself—that represents *the way Cosmo actually is*. But this doesn't give Cosmo a distinctive role *vis-à-vis* actuality, since every Lewis-world represents the way it itself is. $\ulcorner$Actually $\phi\urcorner$ involves indexical reference to Cosmo only when ϕ already involves indexical reference to Cosmo. Since quantification is often tacitly restricted in a broadly "indexical" way, many claims which contain no overt indexical words are indexical all the same. There is no particular reason to think that covert indexical reference to *Cosmo* is particularly common among the folk, however, although it might be more common among certain physicists and philosophers.

So much for what actuality is *not*. But what is it? What does 'actually' *actually* mean? Hazen (1979) was the first to notice the difficulty of extending the counterpart-theoretic translation scheme of CTQML to a language containing an 'actually' operator. Fara and Williamson (2005) explain why some natural ideas, as well as some more intricate proposals due to Graeme Forbes and Murali Ramachandran ***cite***, do not succeed in endowing the 'actually' operator with anything like the right logical behaviour.

So long as we admit the existence of entities playing the standard possible-world role—either by identifying counterpairings with sets of pairs and endorsing Semi-Rigid Pairs, or by rejecting set-theoretic reductionism—this challenge is easily met in the present framework. We can simply analyse $\ulcorner$Actually $\phi\urcorner$ as $\ulcorner$at $w_@$ $(\phi)\urcorner$, where '$w_@$' is a name for the entity corresponding to the plurality consisting of all and only the identity pairs. (In the temporal context, we could do the same for $\ulcorner$Now $\phi\urcorner$.)

One might object to this approach on the grounds that even if there are entities playing the possible world role, their existence and their modal profiles are not *logical* matter, whereas 'actually' is a logical operator, certain principles involving which are logical truths. I am not, myself, much concerned with this kind of objection. But it is interesting to see that there is another approach to the analysis of 'Actually' to which this worry does not

[30] Cf. *OPW* 230: '[A] possible world. . . is a way that an entire world might possibly be. But lesser possible individuals, inhabitants of worlds, proper parts of worlds, are possibilities too. They are ways that something less than an entire world might possibly be.'

apply. In this approach, a sentence ϕ involving multiple 'actually' is first translated into a sentence in the multi-sorted language $\mathcal{L}_W$ introduced in the previous section. Suppose we begin with a plural modal language with the modal operators $\Diamond$, $\Box$, and ACT. Given a formula ϕ of this language, we can transform it into a formula of $\mathcal{L}_W$ by (i) replacing each occurrence of 'ACT' with 'at W_0'; (ii) replacing each occurrence of '$\Diamond$' with '$\exists W_1(PW_1 \wedge \ldots)$'; (iii) replacing each occurrence of '$\Box$' with '$\forall W_1(PW_1 \to \ldots)$'; and (iv) writing '$\exists W_0(AW_0 \wedge \ldots)$'around the resulting expression. We can then apply the method from the previous section to transform this formula of $\mathcal{L}_W$ back into a formula of the modal-operator-free fragment of the original language. It is easy to see that this translation will translate logical truths in the original language into logical truths, so long as we are prepared to treat the existence and uniqueness of ordered pairs as a logical matter.

It is important to understand the difference between the sense in which this theory provides an explanation of sentences involving 'Actually' and the sense in which the counterpart theory of section 4.1 provides an explanation of claims of the form 'Possibly ϕ'. The latter can be viewed as a collection of *analyses*—English sentences of the form 'For it to be possible that ϕ is for it to be the case that . . . '. Our "translations" of claims involving 'actually' are not like this. The translation of 'Actually Mars is red' into $\mathcal{L}_W$ is '$\exists W_0(AW_0 \wedge$ at W_0 Mars is red)'. And the translation of this back into the original language is '$\exists V(\forall x(Vx \leftrightarrow \exists y(x = \langle y, y\rangle)) \wedge \exists x(V\langle \text{Mars}, x\rangle \wedge x \text{ is red}))$'. Given the existence and uniqueness of ordered pairs, this sentence (call it ψ) is equivalent to 'Mars is red'. Since it is not necessary that ((Actually, Mars is red) iff Mars is red), it is not necessary that ((Actually, Mars is red) iff ψ). This seems to rule out our claiming that *for it to be the case that* actually, Mars is red is *for it to be the case that* ψ.

Does this mean that those who take this approach are, after all, forced to accept "primitive modality" in the form of the 'actually' operator? I think not. It is a familiar point that the philosopher's 'actually' is impoverished by comparison with comparable devices in natural language. Actual sentences using 'actually' seem to exhibit something akin to scope ambiguity: it can locally undo some but not all of the modal operators in whose scope it occurs. For example, 'Necessarily, everyone could have had a child he doesn't actually have' is naturally read as inconsistent with the claim that possibly, some parent x has a child y and could not have had any child other than y. This militates in favour of treating these sentences using some formal language more flexible than the standard language of 'actually'. The obvious approach is a language with explicit quantification over worlds, which lets us use the indices on the world variables to keep track of the exact effect we want for a given occurrence of 'actually'. If that is what is going on in the natural language sentences that serve to introduce us to the formal actuality operator, it is far from obvious that the standard logic of that operator is a good guide to the character of the propositions it allows us to express. It would not, in that case, be outrageous to claim that 'Actually, Mars is red' does *not* semantically express a necessarily true proposition. Sentences involving 'actually' might be semantically analogous to sentences involving free variables, in that their compositional semantic contribution cannot reasonably be captured by associating them with a single proposition.

Of course, if we embrace the existence of *entities* (e.g. sets of ordered pairs*) playing the role of possible worlds, there will be no obstacle to our introducing a rigid designator '@' stipulated to refer to whichever one of these entities corresponds to the plurality of all identity pairs, and stipulating that 'Actually' means the same as 'At @'. In this ontological setting, there is no obstacle to our taking the logic of 'actually' fully seriously as a guide to the modal profiles of the propositions expressed. My considered view is that counterpart theorists should, ultimately, want a full-fledged direct analysis of English sentences involving the words 'possible world' and 'instant', as opposed to a reconstructive analysis of sentences in a multi-sorted language. And if they have a direct analysis, they can happily accept the standard view that sentences that begin with 'actually' express non-contingent propositions. I am merely arguing that the case for that standard view is not strong enough to provide an *additional* reason to be dissatisfied with a reconstructive analysis.

4.7 Counterpart theories as A-theories

We are finally ready to answer the question how counterpart theories fit into chapter 1's dichotomy between A- and B-theories. As we noticed in chapter 2, counterpart theorists are not subject to the "primordial objection" to the B-theories. How do they manage this? Do they show that there is something wrong in the conception of the B-theories presupposed by the primordial objection? Or do they manage it by being, in fact, A-theories?

Recall that chapter 1 reconstructed the dispute between temporal A- and B-theorists as turning on questions T1 and T2:

(T1) Are there temporarily true propositions?

(T2) Is the unit class of the present instant much *more natural* than the unit classes of most other instants?

and the dispute between modal A- and B-theorists as turning on questions M1 and M2:

(M1) Are there temporarily true propositions?

(M2) Is the unit class of the actual much *more natural* than the unit classes of most other possible worlds?

It is clear that temporal and modal counterpart theorists should answer 'yes' to T1 and M1 respectively. Given counterpart theory, a temporarily [contingently] true proposition is just a true proposition with false counterparts (i.e. a true proposition that is mapped to a false proposition by some counterpairing). And given Counterpairings, the only way to make true a sentence like

(26) The proposition that Obama is president is such that necessarily [always], it is true iff Obama is president

is to ensure that every counterpairing that maps Obama to a non-president maps the proposition that Obama is president to an untrue proposition. So if sentences like (26) are to be true (on *any* reading!), there must be temporarily [contingently] true propositions.

The status of T1 and M1 in counterpart theory is thus fairly clear. And given what we have learnt about the right way to integrate talk of worlds and instants into counterpart theory, the answers to T2 and M2 should be just as clear. Suppose that we identify worlds/instants with *sets of ordered pairs*. As such sets go, the one consisting of all and only the identity pairs is clearly a very natural one; its unit class is much more natural than the unit classes of most of the other sets of ordered pairs that are worlds or instants. This can be seen as arising from the fact that *identity* is a very natural *relation*, as it clearly is. Admittedly, the identification of worlds or instants with sets of ordered pairs faces a problem: in order to accommodate the entire world role, it requires us to give up the natural principle that ordered pairs have their elements necessarily. But if we are prepared to bite this bullet, the case that the set of identity pairs is especially natural is not affected: the mere fact that the respect in which it is distinctive is a *contingent* one does not diminish its naturalness. Meanwhile, if we find the bullet too big to bite, and consequently reject the identification of worlds or instants with sets of ordered pairs (perhaps in favour of identifying them with sets of ordered pairs*), the case for a positive answer to M2/T2 is no less strong. Somehow or other, the entities that are worlds/instants "correspond" to pluralities of ordered pairs. And whatever our theory of this correspondence relation might be, for the purposes of assessing the naturalness of sets of worlds/instants, we should treat it as a natural relation. When F-ness is a much more natural property than G-ness, and x is some reasonably fundamental entity, we want the set of worlds at which x is F (that is, the set of worlds which correspond to pluralities of pairs that map x to an F thing) to be much more natural than the set of worlds at which x is G (that is, the set of worlds corresponding to pluralities which map x to a G thing). The status of the 'correspondence' relation itself can be factored out when we are thinking in this way, in just the same way that, when we are thinking about the naturalness of sets of ordered pairs, the status of the relation between an ordered pair and its members can be factored out. For this reason, I cannot see that the denial of the naturalness of the correspondence relation provides a tenable route towards a negative answer to M2 or T2.

I conclude that, properly developed, modal counterpart theory is a kind of modal A-theory, and temporal counterpart theory is a kind of temporal A-theory. This conclusion is especially important in the temporal case, since unlike the better-known forms of temporal A-theory, temporal counterpart theory is consistent with a full-fledged endorsement of the manifold picture. The temporal counterpart theorist has no need to posit any suspect additional structure in spacetime beyond what we are used to. Spacetime is broadly homogeneous; no slice through it enjoys any very special distinction. Nevertheless, the present instant does enjoy a very special distinction. This is consistent, because 'the present instant' is not a name for a slice through spacetime: rather, it is a name for a certain set of ordered pairs, or something that in some way corresponds to a plurality of ordered pairs; and it is presentness just amounts to its being a set of identity pairs, or corresponding to all and only the identity pairs.

I expect that many philosophers will resist my claim that temporal counterpart theory is a version of the A-theory. In the literature, temporal counterpart theory has normally been discussed as one of the tenets of 'the stage view', whose other main tenet is the ordinary objects are "stages" (things each of which occupies a single three-dimensional spacetime region). It is common to come across the idea that the difference between the stage view and the competing 'worm view' is merely a "semantic" one, as opposed to a "metaphysical" one. For example, Sider introduces the stage view as follows:

> At one level, I accept the ontology of the worm view. I believe in spacetime worms, since I believe in temporal parts and aggregates of things I believe in. I simply don't think spacetime worms are what we typically call persons, name with proper names, quantify over, etc. The metaphysical view shared by this 'stage view' and the worm view may be called 'four-dimensionalism', and may be stated roughly as the doctrine that temporally extended things divide into temporal parts. (Sider 1996, p. 433)

Here Sider is echoing Lewis, who says something similar when he is considering the analogous contrast in the modal case:

> The theory that ordinary things are trans-world individuals, unified by counterpart relations among their stages, really is a different theory from mine. But the difference is limited. There is no disagreement about what there is; there is no disagreement about the analysis of modality. Rather, there is extensive *semantic* disagreement. It is a disagreement about which of the things my opponent and I both believe in are rightly called persons, or sticks, or stones. (OPW, p. 218)

And many authors have followed Sider and Lewis in emphasising this contrast between shallow "semantics" and deep "metaphysics". If you are used to thinking in this way, and you think that the A-theory/B-theory dispute is a matter of metaphysics (and if *depth* is a hallmark of metaphysics, you definitely should think this this, since the A-theory/B-theory dispute is one of the deepest in philosophy) you will find it hard to believe my conclusion that temporal counterpart theory is a kind of A-theory, and you will think that something must have gone wrong with my argument for that conclusion.

But the idea that any of these disputes are *merely* about 'semantics' cannot be taken seriously. Given uncontroversial principles of disquotational reasoning, any dispute about what a predicate 'F' applies to, or what it takes for a sentence 'S' to be true, will also involve a dispute about the question which things are F, or about what it takes for it to be the case that S; and the latter dispute is (in most cases) not about language at all. Those uncontroversial principles allow us to move back and forth as we please between the material mode and the formal mode; and this is so whether the questions we are interested in are deep and interesting or shallow and boring.

What, then, can it mean to say that the stage theorist and the worm theorist have the same 'metaphysics' or 'ontology'? Presumably the claim is not just that they agree as regards the cardinality of the universe! Perhaps the thought is that they agree as regards *the fundamental structure of reality*. I am not quite sure what that means, but let us grant that it is so. What I don't see is why the A-theory/B-theory dispute needs to be understood as involving different answers to the question 'What is the fundamental structure of reality'? For what would those answers be? If we want to bring the word 'fundamental' into a characterisation of the A-theory/B-theory dispute, the natural thing to say is that A-theorists think that the fundamental facts are *changeable*, whereas B-theorists think that they are not. The dispute need not concern the question how the world *is* structured; the dispute concerns whether that structure is a changing matter.

Orthodox A-theorists, e.g. 'moving spotlight' theorists who enrich a standard spacetime-based fundamental metaphysics with something like a special 'glow of presentness', clearly do disagree with orthodox B-theorists about the fundamental structure of the world. However, we can *imagine* a strange B-theorist who, at some particular instant t, completely agrees with some moving spotlight theorist's account of the fundamental structure of the world. This B-theorist thinks that, in addition to all the structure studied by physics, there is a special extra fundamental property ('glow') that is uniquely instantiated—permanently, of course—by the slice t. (As t has been approaching, this B-theorist has been getting more and more excited.) At t, these characters agree about the specialness of t. Their disagreement is then just over whether this specialness is a changeable or a permanent matter. And this need not manifest itself in any *other* divergence between their accounts of fundamental structure. In particular, the A-theorist might be the kind of A-theorist who accepts biconditionals like 'It will be the case that ϕ iff at some time after the present, ϕ' and 'It has been the case that ϕ iff at some time before the present, ϕ' as *reductive analyses* of the tense operators, and thus sees no need to add posit any special, fundamental 'tense operator structure' in addition to the glow of presentness. (Nor need she take 'at' as primitive: perhaps she analyses 'At t, ϕ' as 'At a world just like the actual world except that t glows, ϕ'.) We can, if we wish, approach this dispute at the linguistic level, in which case we will characterise the A-theorist as claiming, and the B-theorist as denying, that 'glow' features in a certain way in the semantics of tensed discourse. But the possibility of such semantic ascent should not cause us to dismiss the debate as just about words! For *any* philosophical dispute, it is possible to be a "reductionist" about some of the vocabulary in term of which the dispute is framed. If we think the only deep disputes are ones in which both sides agree that all the relevant vocabulary is fundamental, almost no philosophical disputes are deep!

My interest in the question where counterpart theory belongs in the A-theory/B-theory dispute is not motivated merely by classificatory zeal. I am hoping to persuade some people who are attracted to the A-theory that they should be temporal counterpart theorists, on the grounds that this will allow them to reconcile themselves with the manifold picture. For this dialectical purpose, what matters is not so much whether temporal counterpart theory counts as a kind of A-theory, as whether temporal counterpart theory provides whatever it is that attracts these people. So, what does motivate people to be A-theorists? Chapter 2 listed several possible factors—various fancy arguments, as well as a certain sense of rock bottom obviousness that I attempted to channel in discussing 'the primordial objection'. None of these factors does anything to favour the more familiar varieties of A-theory over temporal counterpart theory.

Did I miss something? Well, there may be some A-theorists who find it primitively obvious that *tense operators are fundamental*—facts stated with their aid cannot be "reduced" to any facts statable without tense operators. Those who are gripped by this conviction will be hostile not only to B-theoretic reductions of tense operators, but to the many reductionistic strategies that are consistent with the A-theory as understood in chapter 1. For example, they will resist the view on which 'It will be the case that ϕ' can be reduced to 'ϕ at some future instant', even when this is combined with the claim that futurity is a fundamental, irreducible, monadic property of instants. While they might agree that these claims of these two forms are necessarily and always alike in truth value, it will strike them as obvious that no such necessary equivalence could satisfy whatever further features are required for a "reduction". I find this conviction rather hard to sympathise with: it seems to me that if one is going to go around theorising about which bits of vocabulary are fundamental, one should be open to the possibility that any given ordinary expression fails to be fundamental, although one may legitimately dismiss certain particular hypotheses about *how* a given expression reduces to the fundamental as obviously false. But maybe it is true that some A-theorists are driven by this sort of conviction.[31] If you are one of them, it is not open to you to accept temporal counterpart theory taken as a *reduction* of the tense operators. But I still want to urge you to accept temporal counterpart theory as an extensionally adequate account of the truth values of sentences involving temporal operators, at the actual world and other nomologically possible worlds, if not at all metaphysically possible worlds. For by doing this, you will still be able to make your view consistent with a large and important fragment of the manifold picture. In particular, you will be able to accept *Manifold Naturalness*, the claim that the only natural classes of spacetime points and regions (and sequences and other set-theoretic constructions thereof) are those classes that can be picked out in some reasonably simple way in the language of spacetime physics. Where there is homogeneity as regards the distribution of physical fields, there is homogeneity *simpliciter*. This claim is consistent with the idea that there are all kinds of further facts which do not reduce to the facts statable in terms of that language, or even metaphysically supervene on those facts; what matters is just that these further facts *as a matter of fact*, and perhaps of physical law, do not draw any interesting distinctions between items which are on a par from the standpoint of spacetime physics.

(The only caveat is that for this reconciliation to work, one will have to combine one's temporal counterpart theory with a fairly *simple* analysis of 'future-counterpart' and 'past-counterpart' (or 'future-counterpairing' and 'past-counterpairing') in spacetime-theoretic terms. Otherwise, the fact that this particular relation enjoys the privilege of relating to the facts about how things will be and were, while other similarly simple relations enjoy no such privilege, will constitute a violation of *Manifold Naturalness*. This might or might not be a big constraint. Chapter 6 will consider in more detail the choices one faces in giving a theory about what temporal counterparthood *is*.)

Another factor which may play an important role in motivating people to be A-theorists is the conviction that there are (or have been, or will be) temporary beings—things such that it is not always the case that they are

[31] This might help explain the appeal of speculations about two-dimensional time, which arguably (see section 1.1) arises from some thought to the effect that *nothing* we could say about how things *are*—even if this includes claims about what *times* there are, and how things are at those times, and which of those times have the monadic fundamental properties of pastness, presentness, and futurity!—could ever entail anything about how things *were* or *will be*.

identical to something, unrestrictedly speaking. (This is customarily conceived of as a commitment of "presentism".) As we have seen, this claim is *prima facie* inconsistent with the B-theory—given the standard instant role, the present instant is such that there is nothing that is not, at it, identical to something; if not all instants have this feature, it amounts to the sort of natural distinction for the present instant which B-theorists reject. I am not, myself, tempted to regard it as pre-theoretically obvious that that there are temporary beings in the relevant precise sense; but I will not now try to address the question whether this would be a good starting point for an argument for the A-theory. Suppose that you do find it pre-theoretically obvious that there are temporary beings. Then you cannot accept temporal counterpart theory in the form in which it has been developed in the last two chapters, even as merely extensionally adequate. For if BTCT is extensionally adequate, then the claim that there is something that will fail to be anything must have the same truth value as the claim that there is something that has a future-counterpart that fails to be anything, which is a logical falsehood.

This need not, however, prevent you from accepting the basic strategy for reconciling the A-theory with *Manifold Naturalness*. ***

*** One last motivation: open-futurism.

Many philosophers think of modal counterpart theory as claiming that there is something deeply right about the hyper-essentialist idea that any object which has a property cannot exist without having that property. In this picture, the counterpart theorists' accounts of how sentences of the form 'Possibly, a is F' can be true even when a is not in fact F are conceived of as nothing more than devices for "speaking with the vulgar while thinking with the learned". Given the counterpart theorists' semantics, we cannot say that 'objects have all their properties essentially' is literally true in English, but there may still be something importantly right that can be gestured towards by such formulae. If you thought that counterpart theory had to involve some such commitment, you would surely not regard temporal counterpart theory as doing justice to the truth of the A-theory. For on this conception, temporal counterpart theorists are committed to there being something deeply right in the vicinity of the slogan 'Any object which has a property *never* exists without having that property', despite the fact that the slogan is not literally true. A-theorists are of course likely to be hostile to the thought that there is anything importantly true that could be gestured towards by this slogan: objects really do change, end of story! So it is worth emphasising that temporal counterpart theory as I have developed it is not committed to there being any interesting truth in the vicinity of this slogan; nor does it suggest any obvious candidate for the role of such a truth.

Even those philosophers who do not think of counterpart theorists as "closet hyper-essentialists" often think of counterpart theory as embodying a certain attitude of "metaphysical unseriousness" about *de re* facts. The attitude in question might be expressed using the following metaphor: when God makes the world, all he has to do is decide the answers to the purely qualitative questions. Having decided which qualitative roles are to be played, He does not have to make further decisions about the identity of the particular objects which play the roles. The best-understood way to cash out such metaphors is in modal terms: necessarily, for any *de re* truth, there is some purely qualitative truth that entails it. Call this 'modal anti-haecceitism'. Modal anti-haecceitism is certainly an interesting view, but it is not a commitment of most versions of counterpart theory, including Lewis's original version. Suppose that at a certain world w, and at every world where the purely qualitative facts are just as they are w, x has two counterparts, only one of which is F. Then it is possible that (x is F and there is no purely qualitative truth P such that necessarily, if P, then x is F). And of course modal anti-haecceitism is not a commitment of the modal counterpart theory developed in the last two chapters: on that theory, there are contingent *de re* truths, and no such truth could be entailed by the purely qualitative facts, which are necessary.

However, there is another, much less well-understood, but potentially more interesting way of cashing out metaphors about God's choices: namely, as attempts to articulate *non-factualism* about the *de re*. (For the general idea of non-factualism, see section 1.5.) Non-factualists about the *de re* might claim that while there are *de re* propositions, some of which are true and others of which are false, every PROPOSITION—every proposition which is either TRUE or FALSE—is purely qualitative. Or they might claim that whenever it is *factually* the case that P, the purely qualitative facts entail that P. The idea of the metaphor is that God only has to concern himself with settling the TRUTH-VALUES of PROPOSITIONS, or with settling what is factually the case. Small-p propositions get to be true or false without his needing to bother about them.

Non-factualism about the *de re* is an extreme view: even those who are comfortable with non-factualism as a general theoretical option might regard it as too outlandish to be taken seriously. When we are introducing non-factualist views about other domains, the relevant notion of factuality is often introduced with the help of paradigms, which are often *de re* facts: the Earth is round; this is a table; there is a tree in the Quad. A view where none of these paradigms is factual pushes our understanding of 'non-factual' very far, perhaps to the breaking point. Moreover, non-factualism about the *de re* is hard to assimilate to the best-developed non-factualist projects. Based on the example of expressivism in ethics, it is natural to think that non-factualists about a certain subject matter face the burden of telling some revealing story about the true nature of the states that we normally call *beliefs* about the relevant subject matter, where the story in question has to be told in factual language. But when p is a *de re* proposition, *believing* p is a *de re* property—if we are not modal anti-haecceitists, we should not expect to be able to state necessary and sufficient conditions in purely qualitative terms for, e.g., believing Obama to be tall. It is thus not clear that there is any body of theory we could develop that would flesh out non-factualism about the *de re* in anything like the way in which substantive expressivist accounts of the states normally known as 'moral beliefs' flesh out non-factualism about morality.

There is very little basis for attributing non-factualism about the *de re* to Lewis. Lewis does not seem to be hospitable to the ideology required to get non-factualist views onto the table in any domain. In Lewis's discussions of the areas where non-factualist views have been most prominent—ethics, chance, laws of nature—he does not even mention these views as options; and in the one explicit discussion of non-factualism that I know of (Lewis 2005) is quite unsympathetic. Moreover, as we saw in chapter 3, there is a strong current in Lewis's work that pushes him towards the view that all contingency is *de re* contingency. The combination of this claim with non-factualism about the *de re* yields non-factualism about the contingent. And it would be a real stretch to hang *that* on Lewis. For Lewis, beliefs about the contingent are the paradigm case of belief—indeed, his theory of belief has a lot of trouble dealing with the apparent fact that not everyone believes every necessary proposition. If anything, one would be more tempted to think of Lewis as a non-factualist about the *non*-contingent, insofar as he is drawn to a coarse-grained theory of belief. On such a view, 'Necessarily ϕ' entails 'Everyone believes that ϕ' on the most basic, metaphysically perspicuous way of understanding 'believes that ϕ'. If 'Everyone believes that ϕ' is false on some more ordinary interpretation, then the property expressed on that interpretation by 'believes that ϕ' is not perspicuously understood as that of believing a certain proposition, just as, according to expressivists about morality, the property we normally express using 'believes that lying is wrong' is not perspicuously understood as that of believing a certain proposition.

It is more tempting to attribute non-factualism about the *de re* to those philosophers who think of themselves as counterpart theorists but do not endorse anything like Lewisian modal realism. It is characteristic of these philosophers that while they talk about "possible worlds", they do so in a way that departs from the standard possible world role laid out in section 1.3. For example, they may sometimes affirm there is exactly one possible world at which ϕ (where ϕ is purely qualitative), while also claiming that for some x, it is possible that (ϕ and x is F), and also possible that (ϕ and x is not F), which is incoherent given the standard role. In interpreting such philosophers, we face the question why they use 'world' in this non-standard way, rather than using it to talk about some entities which *do* play the standard world role. Are they just being perverse? Perhaps in some cases the answer is that they don't believe that there *are* any entities that play the standard world role. But most of the time this is unlikely to be the full story, since insofar as they believe in set theory, they should be able to use it to construct entities that do a better job of fitting the standard role than the ones they call 'possible worlds'. The hypothesis that the philosophers in question are non-factualists about the *de re* helps to explain why they should care about the entities they call 'possible worlds', even when they believe in other entities that play a fuller version of the possible world role. For it is characteristic of possible worlds, on the usual view, that each of them determines a natural mapping from propositions to truth-values. A non-factualist about certain subject matters would find it natural to focus for many purposes on entities each of which determines a natural mapping from PROPOSITIONS to TRUTH-VALUES. Such entities will play the possible world role so long as we confine ourselves to factual discourse, although they may not be fine-grained enough to continue playing the role after we have expanded our language to include the relevant non-factual vocabulary. A non-factualist about *de re* propositions would thus find it natural to focus on entities that play the possible world role as conceived by the non-modal-realist counterpart theorists, which coincides with the usual possible world role for purely qualitative sentences. This helps make sense of what would otherwise seem an uninteresting and pointless departure from the accepted use of ‘possible world’.

The combination of non-factualism about the *de re* with the thesis that all change is *de re* change yields non-factualism about the changeable: PROPOSITIONS never change their TRUTH-VALUES; whenever it is factually the case that *P*, it is always the case that *P*. Likewise, the combination of non-factualism about the *de re* with the thesis that all contingency is *de re* contingency yields non-factualism about the contingent.

Chapter 1 discussed non-factualism about the changeable and non-factualism about the contingent as possible strategies for resisting the argument from propositional temporalism [contingentism] to temporal [modal] elitism. It is plausible that if one recognises a category of non-factual truths, one should not take these truths into account in the same way as factual truths when one is assessing the naturalness of a given class of objects. For example, non-factualists about morality will presumably not want to say that the class of morally wrong actions is a very natural one, despite the fact that it is the extension of the very easy-to-refer-to property *being morally wrong*. The facts about the naturalness of classes are supposed to capture the structure of reality: and this is supposed to be a paradigmatically factual subject matter. The application of this idea to non-factualism about the *de re* is not entirely straightforward. Non-factualists about the *de re* will not agree that *all* truths about naturalness are factual truths—*de re* truths about naturalness, such as the truth that a certain particular class is very natural, will share in the non-factual character of all *de re* truths. It is only the purely qualitative questions about naturalness—such as the question ‘Is there an instant whose unit class is much more natural than those of most other instants?’—whose answers are factual. Because of this, it is far from clear how non-factualists about the *de re* should go about forming opinions about the naturalness of classes on the basis of their other opinions. It is clear enough, on the other hand, that non-factualists about the changeable [contingent] should take only permanent [necessary] properties into account in assessing the naturalness of a class. Since non-factualists about the *de re* who also think that all change [contingency] is *de re* change [contingency] must be non-factualists about the changeable [contingent], such philosophers have principled grounds for rejecting the present section’s argument from temporal [modal] counterpart theory to for temporal [modal] elitism.

Non-factualism about the *de re* is an interesting view to think about. But the main point I want to make about it is the simple one that it is certainly *not* a commitment of temporal or modal counterpart theory in general, or of the particular versions of these theories which the last two chapters have developed. And I am emphasising this because I expect temporal A-theorists to resist my attempts to convince them that they should be temporal counterpart theorists, and I suspect that a lot of this resistance will stem from the mistaken assumption that temporal counterpart theory would have to involve non-factualism about the *de re*, or at least some attitude of metaphysical unseriousness about *de re* facts. It is true that a temporal counterpart theory that was coupled with this kind of non-factualism would not give A-theorists what they want, since this combination of views entails non-factualism about the changeable. But there is no reason for a temporal counterpart theorist to feel any sympathy at all for non-factualism about the *de re*; the kind of facts that change according to temporal countepart theorists, namely *de re* facts, can be every bit as factual and robust and fundamental as they seem to be.